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Applied Bayesian Data Analysis in Requirements Engineering

A Gentle Introduction to the Analysis of Quantitative Data within a
Bayesian Framework



17-21 August 2026
Montreal, Canada

Monday, 17th of August, 13:45-17:15 in Room E-2021



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Applied Bayesian Data Analysis

In Requirements Engineering



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JulianFrattini/**bda4sci**

Public

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<http://www.robertfeldt.net/>



Hans-Martin Heyn

Senior Lecturer

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Postdoctoral Researcher

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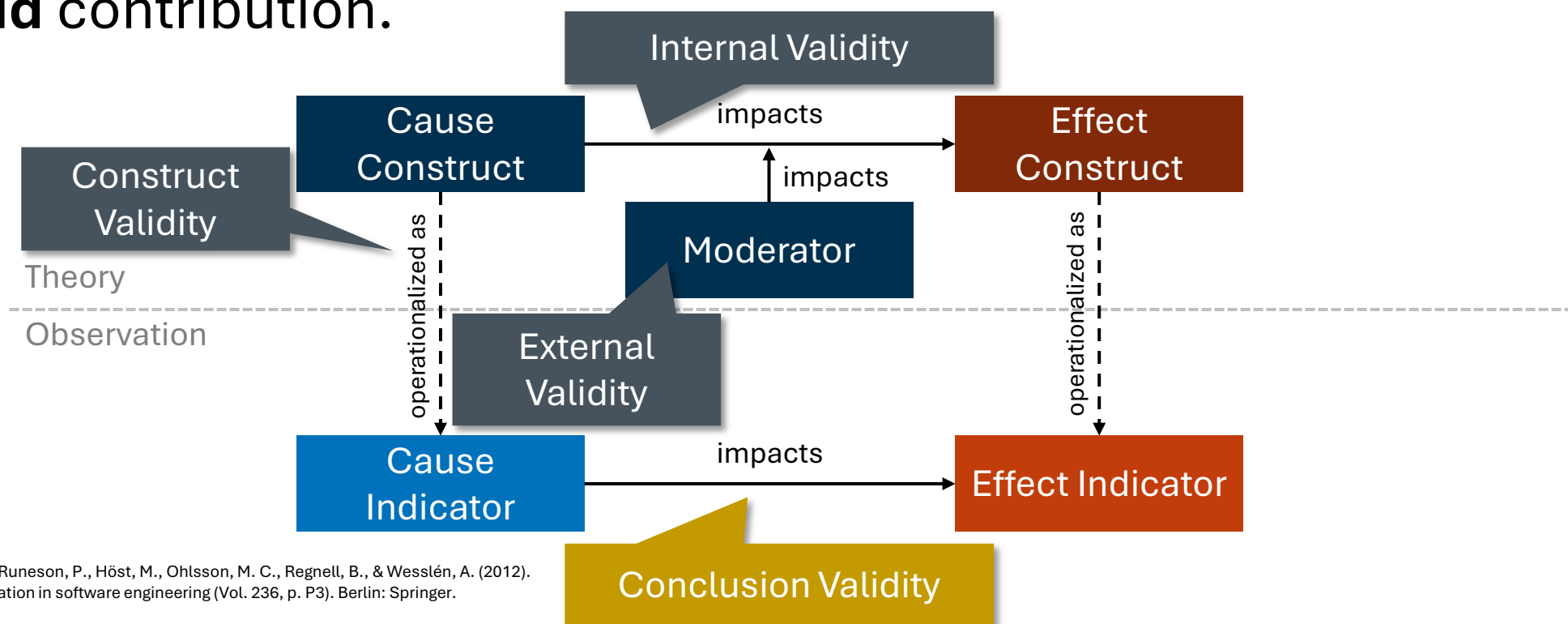
Schedule

Session	Time	Content
Motivation	13:45-14:00	Motivation and learning goals
State-of-Practice	14:00-14:30	Frequentism and its issues
Bayes' Theorem	14:30-15:15	Fundamentals of Bayes' theorem
Break	15:15-17:45	
Bayesian Workflow I	15:45-16:30	Fundamental Bayesian workflow with simple normal data
Bayesian Workflow II	16:30-17:00	Advanced Bayesian workflow with non-normal data
Learning Path	17:00-17:15	Advice for the Bayesian journey
Feedback	17:15-17:45	Reflection, discussion, and closing

Motivation and Goals

Good Science

If we strive to *do good science*, we aim for **rigorous** and **relevant** empirical research contributions, where a rigorous contribution is a **valid** contribution.



Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236, p. P3). Berlin: Springer.

Conclusion Validity

Even if we have ensured the internal, external, and construct validity of our study in the *data collection phase*, we might still draw the wrong conclusions from it.

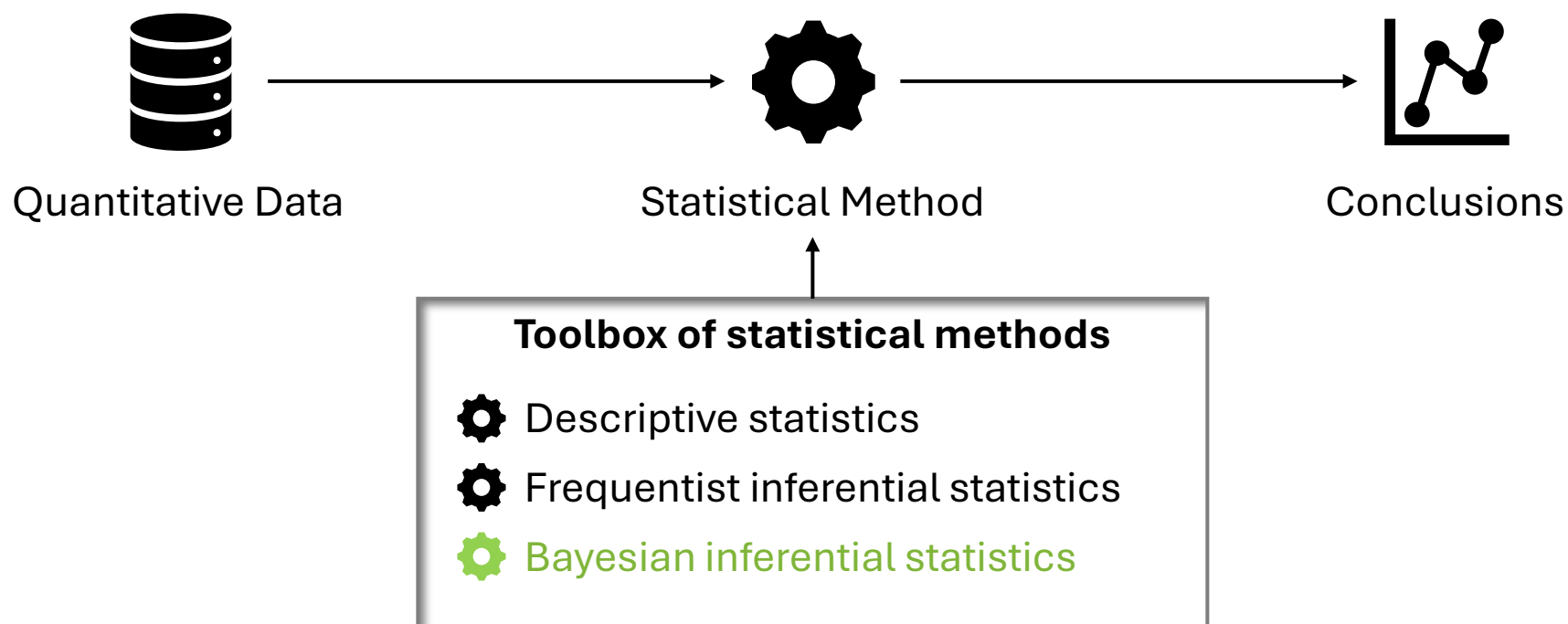


Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236, p. P3). Berlin: Springer.

Conclusion validity affects our “ability to draw the correct conclusion about relations between the treatment and the outcome of an experiment.”

Conclusion Validity

The conclusion validity is determined by the **statistical methods** researchers apply to derive conclusions from our data.



Learning Outcomes

After this tutorial, you will be able to

1. Better understand the current state-of-practice in data analysis and its problems,
2. Understand Bayes' theorem and the Bayesian perspective on probability, and
3. Apply a systematic Bayesian data analysis process.

State of Practice

Review of frequentist methods and their threats to conclusion validity

Null-Hypothesis Significance Tests

The state-of-practice for analyzing quantitative data in SE research focuses on Null-Hypothesis Significance Testing (NHST) commonly consisting of three components.

?

Null-hypothesis

Null-hypotheses are usually formulated such that they encode *no effect*, e.g., “ H_0 : The use of GenAI does not affect a developer’s productivity.”

Null-Hypothesis Significance Tests

The state-of-practice for analyzing quantitative data in SE research focuses on Null-Hypothesis Significance Testing (NHST) commonly consisting of three components.



Null-hypothesis



Data

GenAI Usage	Productivity
10%	3.5
55%	4.2
30%	4.0
...	...

Null-Hypothesis Significance Tests

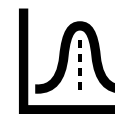
The state-of-practice for analyzing quantitative data in SE research focuses on Null-Hypothesis Significance Testing (NHST) commonly consisting of three components.



Null-hypothesis



Data

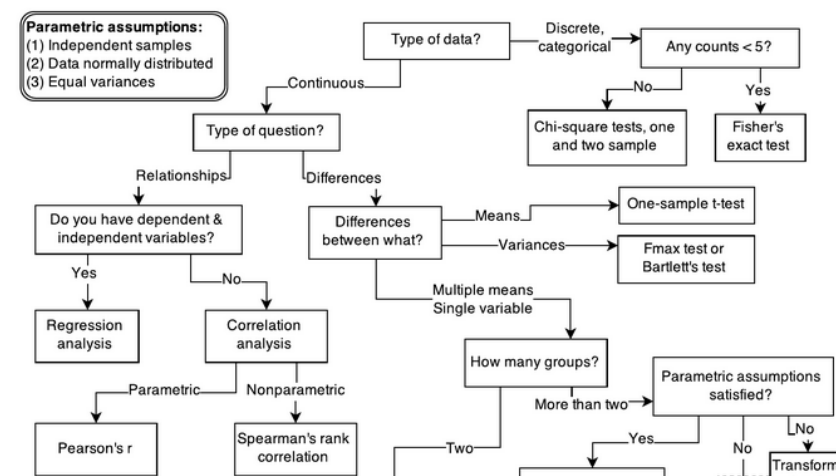


Appropriate test

McElreath, Richard. *Statistical rethinking: A Bayesian course with examples in R and Stan*. Chapman and Hall/CRC, 2018.

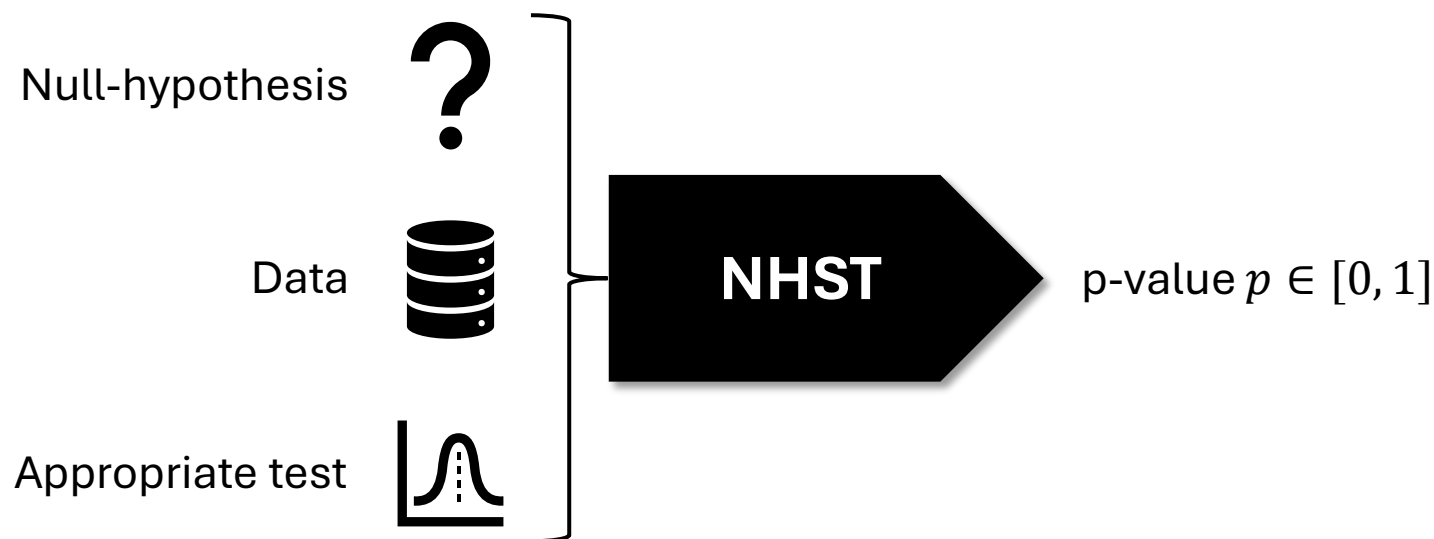
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Bayesian Data Analysis Tutorial



Null-Hypothesis Significance Tests

Null-hypothesis significance tests (NHSTs) work roughly as follows:



What exactly is this mysterious

p-value?

Null-Hypothesis Significance Tests

A p-value is a statement about the observed data.

” A p-value is the probability of observing data **at least as extreme as what you actually got**, *assuming the null hypothesis is true*.

In other words: A null-hypothesis (describing “no effect”) receiving a p-value of 4% on observed data implies that – if that null-hypothesis were true and there is indeed no effect – there is a 4% chance we would observe the collected data.

Null-Hypothesis Significance Tests

A p-value is a **not** statement about the formulated hypothesis.

” A p-value is **not** the probability that the null hypothesis is correct, given what they observed.

However, this is usually the question of interest: how likely is there no/an effect of one variable on another? **NHSTs do not answer this question.**

Null-Hypothesis Significance Tests

To make a decision based on an NHST, the resulting p-value is compared with a significance level α :

- $p < \alpha$ represents a *statistically significant* result
- $p \geq \alpha$ represents *no* statistically significant result

Statistically significant results imply that two distributions are **statistically unlikely** to have come from the same population.

Origin of the NHST

The Null-Hypothesis Significance Test is actually an improper hybrid of two separate techniques:

Significance Tests (by R. A. Fischer)

- Tests a single null hypothesis h_0 against no alternative
- Produces a p-value
- Aims to quantify evidence *against* h_0
- There is no fixed decision threshold (p-values are treated continuously)

Hypothesis Test (by Neyman and Pearson)

- Tests null hypothesis h_0 against an alternative hypothesis h_1
- Defines a decision rule in advance to reject h_0 based on a significance level α and a power β
- Aims to control errors in the long run across repeated decisions

Null-Hypothesis Significance Tests

Analyzing data using only NHSTs has several issues:



Binary

NHSTs reduce complex data to binary, seemingly certain results



Normal

NHSTs distinguish only normal from non-normal data



Interpretability

The p-value is often misinterpreted



Arbitrary

There is nothing special about the significance level $\alpha = 5\%$



The Bayesian Paradigm

Paradigm of Probability

Next to the **frequentist** paradigm, the **Bayesian** paradigm of statistics assigns a different meaning to *probability*:

Frequentist Probability

The proportion of times it would occur if you repeated the experiment an infinite number of times under identical conditions.

→ Probability applies to **events**

"What is the probability of observing productivity data at least as extreme as ours, *assuming GenAI usage has no effect?*"

Bayesian Probability

A statement about *the state of knowledge of an observer that* quantifies how confident a rational agent is that something is true, given their current information.

→ Probability applies to **hypotheses**

"What is the probability that the hypothesis 'GenAI usage improves productivity' is true, given our data?"

Bayes' Theorem

The Bayesian paradigm is based on **Bayes' Theorem**:

$$P(h|d) = \frac{P(d|h) \cdot P(h)}{P(d)}$$

Bayes' Theorem

The Bayesian paradigm is based on **Bayes' Theorem**:

Posterior
Probability that the
hypothesis h is true given
the observed data d

$$P(h|d) = \frac{P(d|h) \cdot P(h)}{P(d)}$$

Bayes' Theorem

The Bayesian paradigm is based on **Bayes' Theorem**:

Likelihood

of observing data d if the
hypothesis h were true

Posterior

Probability that the
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Bayes' Theorem

The Bayesian paradigm is based on **Bayes' Theorem**:

	Likelihood	Prior
	of observing data d if the hypothesis h were true	probability that hypothesis h is true given no information
Posterior Probability that the hypothesis h is true given the observed data d	$P(h d) = \frac{P(d h) \cdot P(h)}{P(d)}$	

Bayes' Theorem

The Bayesian paradigm is based on **Bayes' Theorem**:

	Likelihood	Prior
	of observing data d if the hypothesis h were true	probability that hypothesis h is true given no information
Posterior		
Probability that the hypothesis h is true given the observed data d		
	$P(h d) = \frac{P(d h) \cdot P(h)}{P(d)}$	
	Marginal likelihood	
	i.e., the probability to observe the data d under all possible hypotheses	

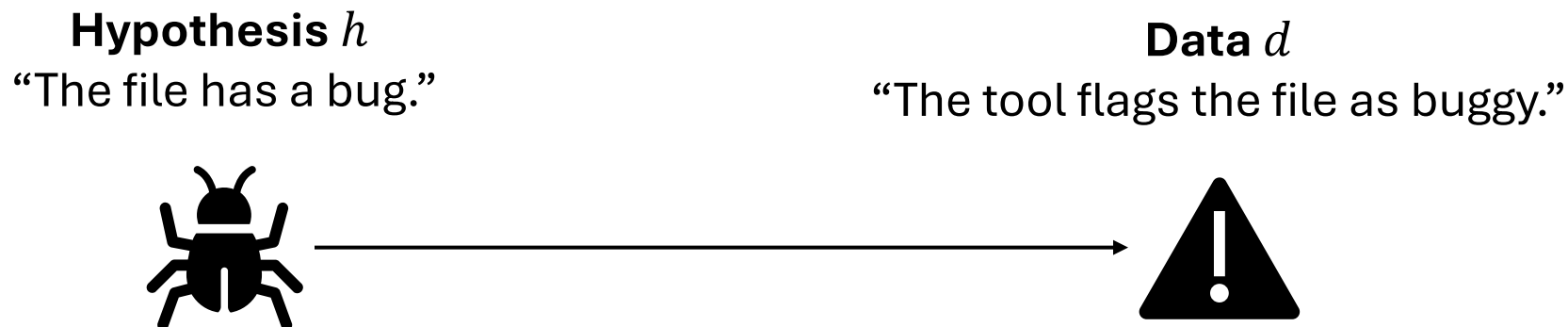
Example

We are investigating a static code analysis tool. The tool has 90% sensitivity¹ and 10% false positive rate. Only 5% of files actually have bugs. *What's the probability that a flagged file is buggy?*

¹ sensitivity = precision, i.e., $TP/(TP+FP)$

Example

We are investigating a static code analysis tool. The tool has 90% sensitivity and 10% false positive rate. Only 5% of files actually have bugs. *What's the probability that a flagged file is buggy?*



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Likelihood

of observing data d if the hypothesis h were true

Prior

probability that hypothesis h is true given no information

Posterior

Probability that the hypothesis h is true given the observed data d

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We are investigating a static code analysis tool. The tool has **90% sensitivity and 10% false positive rate**. Only **5% of files actually have bugs**. *What's the probability that a flagged file is buggy?*

Likelihood **Prior**

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Posterior

Probability that the hypothesis h is true given the observed data d

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Marginal likelihood

i.e., the probability to observe the data d under all possible hypotheses

Example

We are investigating a static code analysis tool. The tool has 90% sensitivity and 10% false positive rate. Only 5% of files actually have bugs. *What's the probability that a flagged file is buggy?*

The **marginal likelihood** $P(d)$ can be calculated from the other data:

$$\begin{aligned} P(d) &= P(d|h) \cdot P(h) + P(d|\neg h) \cdot P(\neg h) \\ &= 0.9 \cdot 0.05 + 0.1 \cdot 0.95 = 0.14 \end{aligned}$$

Marginal likelihood

i.e., the probability to observe the data d under all possible hypotheses

Example

We are investigating a static code analysis tool. The tool has **90% sensitivity and 10% false positive rate**. Only **5% of files actually have bugs**. *What's the probability that a flagged file is buggy?*

Likelihood

of observing data d if the hypothesis h were true

Prior

probability that hypothesis h is true given no information

Posterior

Probability that the hypothesis h is true given the observed data d

$$P(h|d) = \frac{0.90 \cdot 0.05}{0.14}$$

Marginal likelihood

i.e., the probability to observe the data d under all possible hypotheses

Example

We are investigating a static code analysis tool. The tool has **90% sensitivity and 10% false positive rate**. Only **5% of files actually have bugs**. *What's the probability that a flagged file is buggy?*

Likelihood

of observing data d if the hypothesis h were true

Prior

probability that hypothesis h is true given no information

Posterior

Probability that the hypothesis h is true given the observed data d

$$P(h|d) = \frac{0.90 \cdot 0.05}{0.14} = 0.321$$

Marginal likelihood

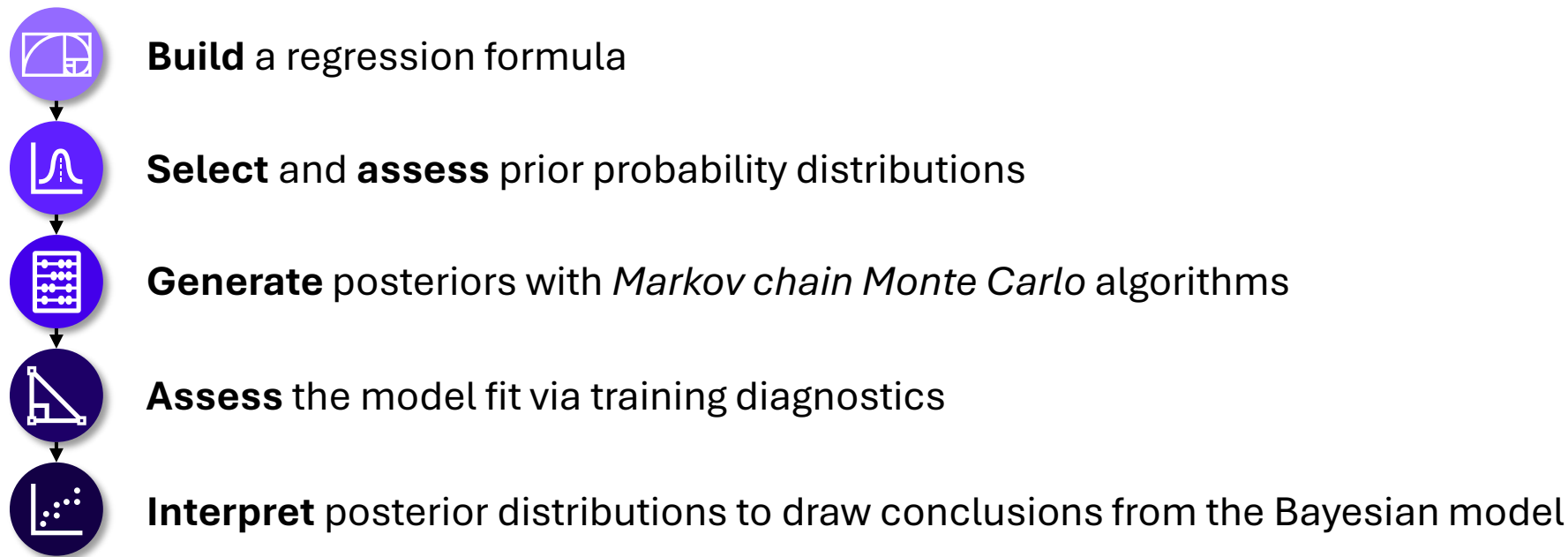
i.e., the probability to observe the data d under all possible hypotheses

Application I

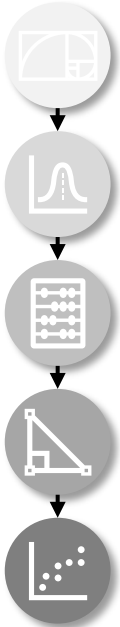
Using Bayesian Data Analysis in Practice

The Bayesian Workflow

The upcoming demonstration will teach you to *understand* or **do** the following steps of a practical Bayesian workflow:



Code Snippets

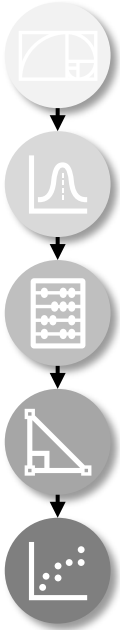


The process will be illustrated both on a theoretical, mathematical and on a practical, code level.

All code-level snippets will be written in R.

```
R 4.5.0  
library(brms)  
  
d <- read_csv("data_file.csv")
```

Precursor

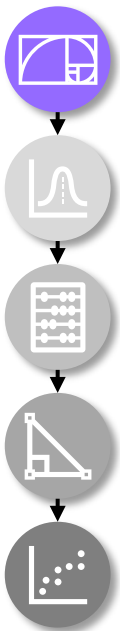


Prior to the now presented workflow, we assume that

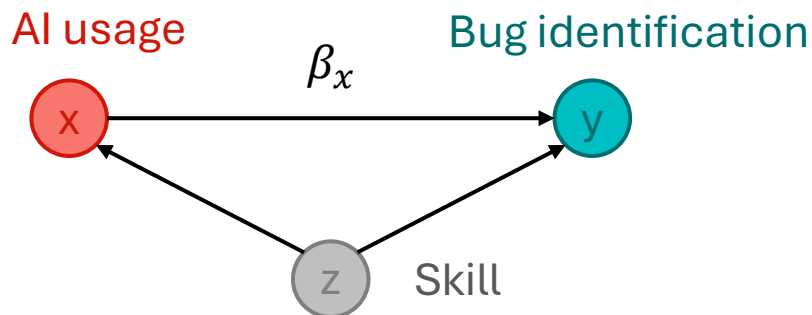
1. we have **modelled our causal assumptions**, e.g., in a directed, acyclic graph (DAG), and
2. we have **collected data** on all relevant variables.

If you participated in last year's tutorial on **statistical causal inference**, this here is the interface.

Context

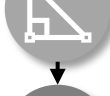
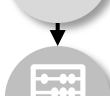
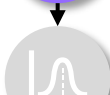


Research question: Does **increased AI usage** affect the **ability to identify bugs** (adjusting for reviewer skill)?

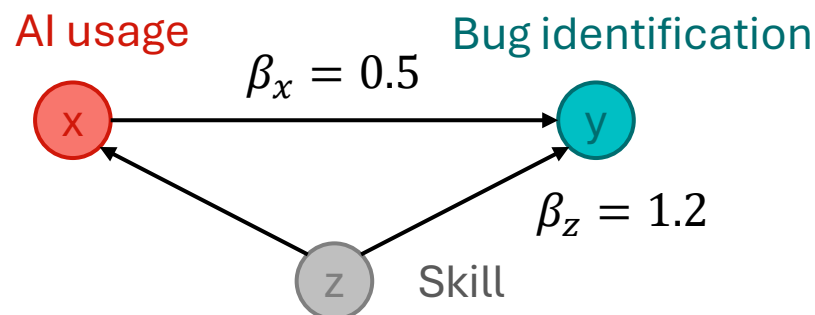


Var	Name	Type
x	Measure of AI usage	Continuous, Gaussian-distributed
y	Ability of identifying bugs in a code snippet	Continuous, Gaussian-distributed
z	Skill of a code reviewer	Continuous, Gaussian-distributed

Context

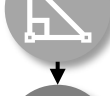
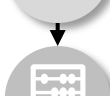
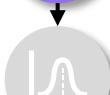


For pedagogical purposes, we **simulate a data set** with known effect strengths.



With our Bayesian data analysis, we aim to recover these effect strengths.

Building a Regression Formula



To determine the target quantity β_x , we can describe the relationship between variables in a regression formula.

Observations about the outcome variable y ...

The shape defining parameter μ , the mean of the distribution, ...



Bayesian Model

$$y \sim N(\mu, \sigma)$$

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

... follow a normal distribution N defined by parameters μ and σ .

... emerges from the sum of several factors.

Building a Regression Formula

To determine the target quantity β_x , we can describe the relationship between variables in a regression formula.

The shape defining parameter μ , the mean of the distribution, ...

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

... emerges from the sum of several factors.

 **Intercept**

represents the likely “baseline” values for μ , i.e., the mean of y without the influence of any other variables like x or z .

 **Terms**

consisting of a combination of

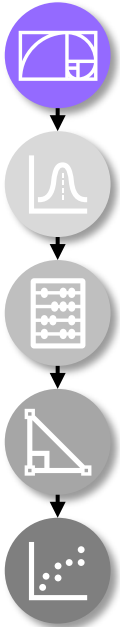
 **Coefficient**

represents the effect that a variable has on the parameter μ , i.e., the change in μ when changing the variable by one unit.

$\times$  **Data**

are the actual values collected for the variables x and z .

Building a Regression Formula



To determine the target quantity β_e , we can describe the relationship between variables in a regression formula.



Bayesian Model

$$y \sim N(\mu, \sigma)$$

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

 **Intercept**

 **Terms**

 **Coefficient**

 **Data**

```
R 4.5.0 using library(brms)
```

```
model <- brm(
  y ~ 1 + x + z,
  family = Gaussian
)
```

Selecting Priors

Every intercept and path coefficient is a **probability distribution** that we need to initialize with domain knowledge.



Bayesian Model

$$y \sim N(\mu, \sigma)$$

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

Selecting Priors

Every intercept and path coefficient is a **probability distribution** that we need to initialize with domain knowledge.

“On average, the value of μ (i.e., the mean of y) is around 3 with a standard deviation of 1.”



Bayesian Model

$$y \sim N(\mu, \sigma)$$

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

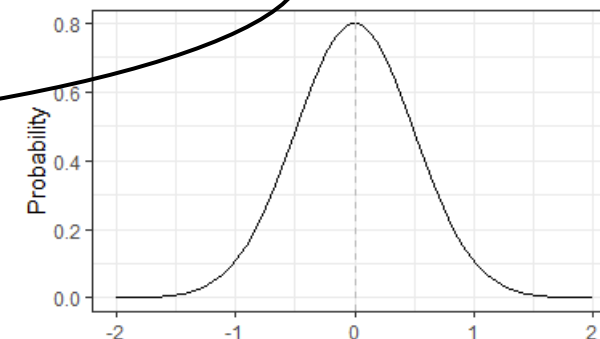
$$\alpha \sim N(3, 1)$$

$$\beta_x \sim N(0, 0.5)$$

$$\beta_z \sim N(0, 0.5)$$

$$\sigma \sim \exp(1)$$

“Before seeing any data, we assume that x can either have a positive, negative, or no effect on μ .”



Selecting Priors

Every intercept and path coefficient is a **probability distribution** that we need to initialize with domain knowledge.



Bayesian Model

$$y \sim N(\mu, \sigma)$$

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

$$\alpha \sim N(3, 1)$$

$$\beta_x \sim N(0, 0.5)$$

$$\beta_z \sim N(0, 0.5)$$

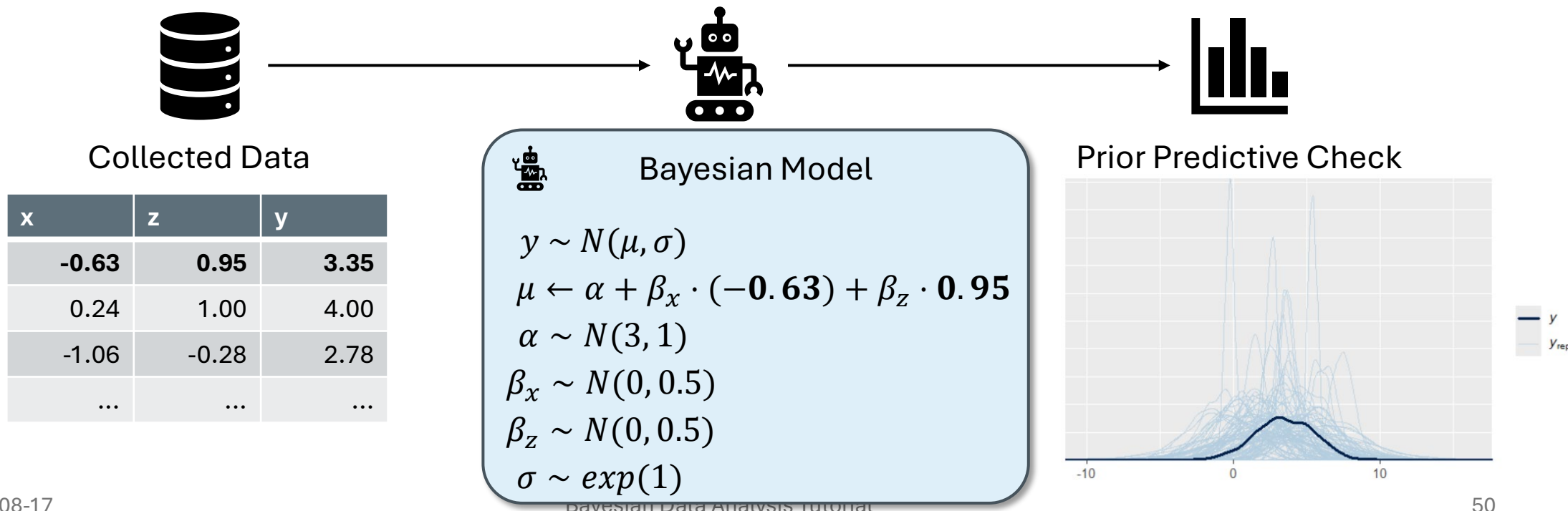
$$\sigma \sim \exp(1)$$

R 4.5.0 using library(brms)

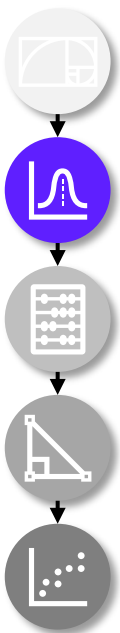
```
model <- brm(
  y ~ 1 + x + z,
  family = Gaussian,
  priors = c(
    prior(normal(3, 1), class = Intercept),
    prior(normal(0, 0.5), class = b),
    prior(exp(1), class = sigma)
  )
)
```

Assessing Priors

The actually observed data should be **plausible** given the prior probability distributions. We can assess this via a **prior predictive check**.



Assessing Priors



The actually observed data should be **plausible** given the prior probability distributions. We can assess this via a **prior predictive check**.

Force the model to sample from the priors directly

```
R 4.5.0 using library(brms)

model <- brm(
  y ~ 1 + x + z,
  family = Gaussian,
  priors = c(
    prior(normal(3, 1), class = Intercept),
    prior(normal(0, 0.5), class = b),
    prior(exp(1), class = sigma)),
  sample_prior = "only",
  iter = 4000, warmup = 1000, chains = 4,
  cores = 4, seed = 4
)
```

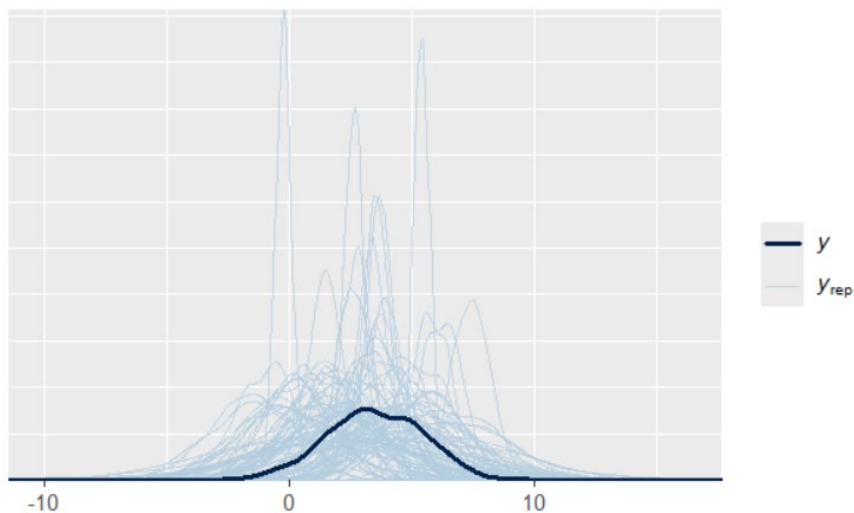
Additional configuration of the model (not important here)

Assessing Priors

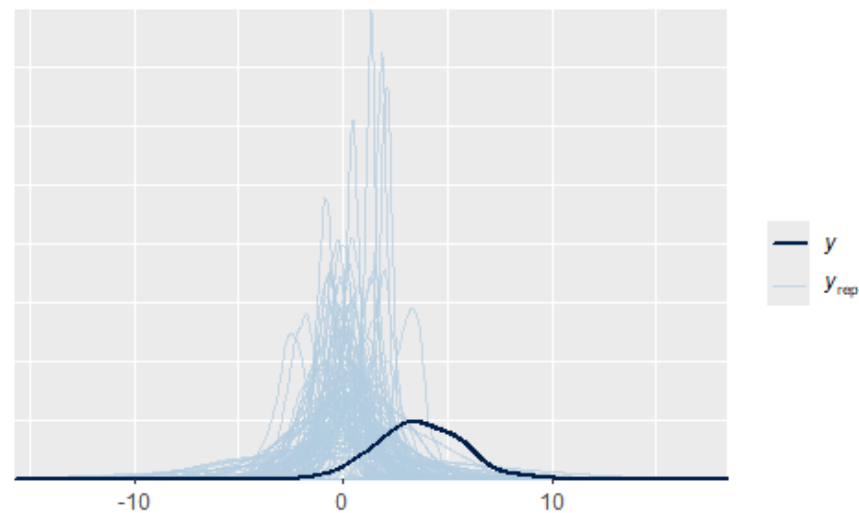
A successful prior probability check shows that sampling from the priors yields data **encompassing** the actual data while **leaving variance around it**.



Successful prior predictive check with
`prior(normal(3, 1), class = Intercept)`

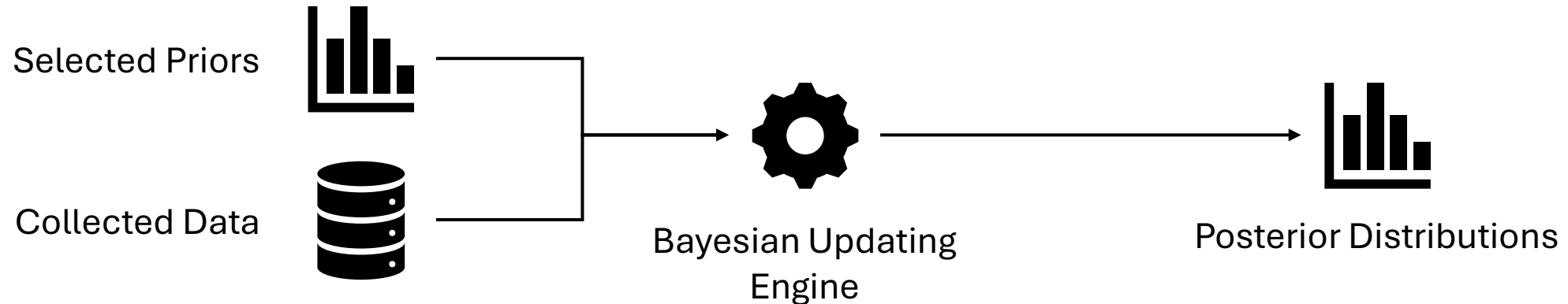


Unsuccessful prior predictive check with
`prior(normal(0, 1), class = Intercept)`



Bayesian Updating

With the prior probability distributions confirmed as plausible, we can perform the actual model training.



At the end of this process, all probability distributions ($\alpha, \beta_e, \dots$) will be updated to **better reflect the relationships** between the variables. This way, we **estimate the effects** between variables.

Bayesian Updating

The model training is based on Bayes' Theorem.

Selected Priors



Collected Data



Bayesian Updating
Engine

Posterior Distributions

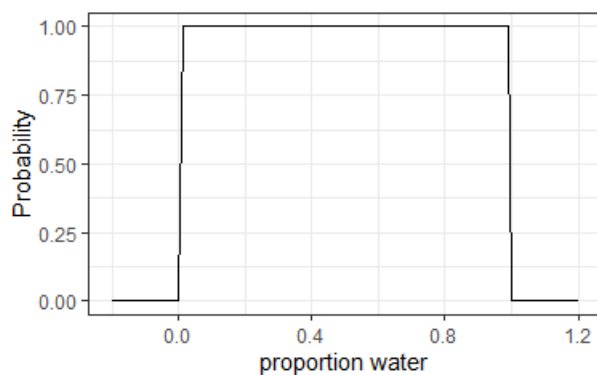


$$P(h|d) = \frac{P(d|h) \cdot P(h)}{P(d)}$$

Bayesian Updating Engine

Imagine estimating the proportion of water on an unknown planet with zero prior knowledge. Every data point is a random probe from the planet's surface, reporting either **Water** or **Land**.

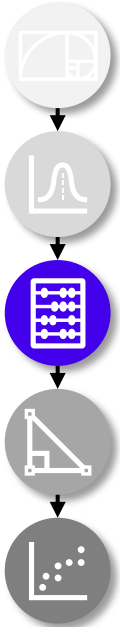
Prior Probability Distribution



Collected Data

{**W**, **L**, **W**, **W**, **W**, **L**, **W**, **L**, **W**}

Bayesian Updating Engine



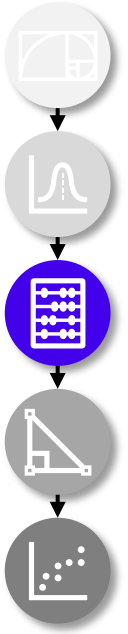
The computational challenge with this process is estimating the **Marginal likelihood** $P(d)$.

$$P(h|d) = \frac{P(d|h) \cdot P(h)}{P(d)}$$

When not dividing the product $P(d|h) \cdot P(h)$ by the marginal likelihood $P(d)$, it **will still be proportional** ($\propto$) but not be a probability distribution bounded within $[0, 1]$ and we cannot update it further after one round.

$$P(h|d) \propto P(d|h) \cdot P(h)$$

Bayesian Updating Engine

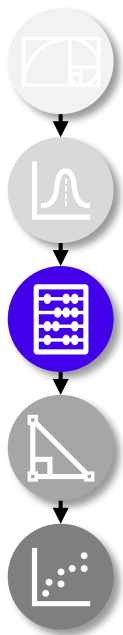


Estimating the **marginal likelihood** $P(d)$ for **binary hypotheses** $H = \{h_0, h_1\}$, this was simply

$$P(d) = P(d|h_0) \cdot P(h_0) + P(d|h_1) \cdot P(h_1)$$

But now our hypotheses are **continuous**, requiring to integrate over a complex parameter space. These integrals often have no closed form.

Markov chain Monte Carlo Sampling

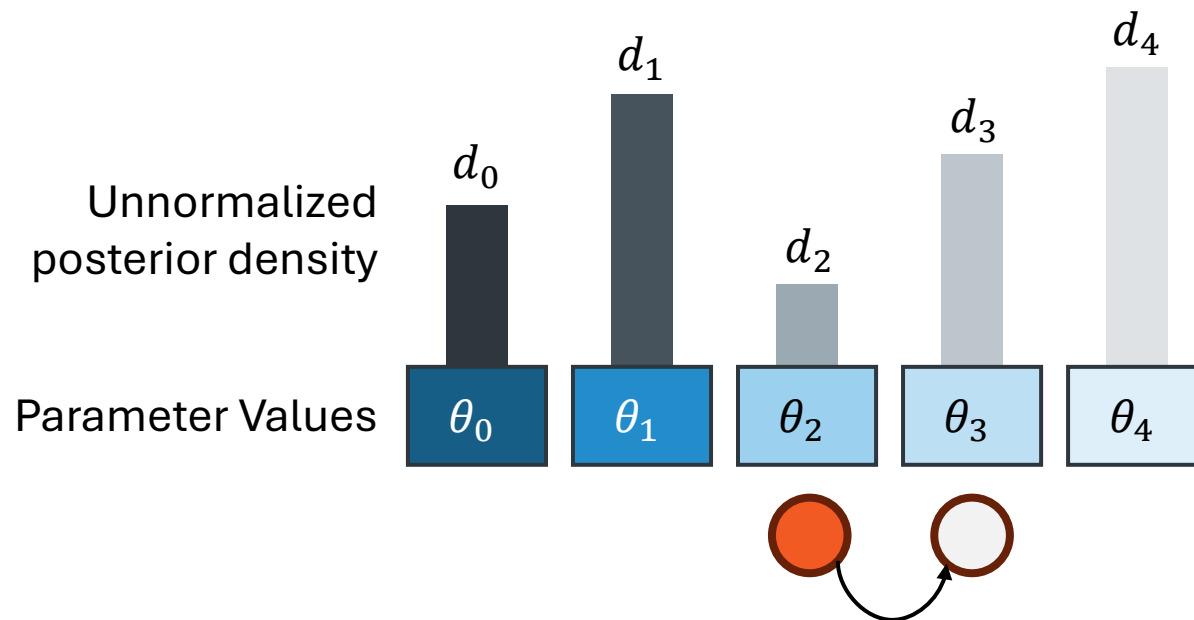


We can sidestep computing the posterior directly by **drawing samples from the posterior** without knowing its normalizing constant. Sampling proportionally to the *relative* posterior density approximates the true posterior.

This can be realized via Markov chain algorithms, e.g., Markov chain Monte Carlo sampling.

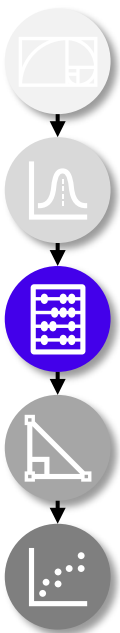
Markov chain Monte Carlo Sampling

Markov chain Monte Carlo samplers travel between adjacent parameter values based on their **relative density**:

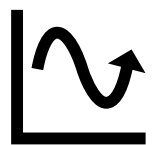


1. Initialize a Markov chain at random x
2. Randomly decide in which direction to move from the starting point
3. Move to the adjacent value with probability $p = \frac{d_{x+1}}{d_x}$ (or $\frac{d_{x-1}}{d_x}$)
4. Repeat from (2)

Markov chain Monte Carlo Sampling



Over time, the chain wanders through the parameter space drawing more samples from high-density regions. Since only the **ratio of densities** matters at each step, the (unknown) normalizing constant becomes irrelevant.



Visualization: <https://chi-feng.github.io/mcmc-demo/app.html>

This is also the reason why BDA has only been adopted so recently: these MCMC sampling algorithms were computationally infeasible just until recently.

Markov chain Monte Carlo Sampling

Executing the MCMC sampler is similar to the prior predictive check, simply removing the `sample_prior = "only"` configuration.

```
R 4.5.0 using library(brms)

model <- brm(
  y ~ 1 + x + z,
  family = Gaussian,
  priors = c(
    prior(normal(3, 1), class = Intercept),
    prior(normal(0, 0.5), class = b),
    prior(exp(1), class = sigma)),
  iter = 4000, warmup = 1000, chains = 4,
  cores = 4, seed = 4
)
```

Click the Refresh button to see progress of the chains
 starting worker pid=26236 on localhost:11598 at
 23:37:14.308
 starting worker pid=33988 on localhost:11598 at
 23:37:14.594
 starting worker pid=12996 on localhost:11598 at
 23:37:14.796
 starting worker pid=23500 on localhost:11598 at
 23:37:14.991

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
 Chain 1:
 Chain 1: Gradient evaluation took 3.5e-05 seconds
 Chain 1: 1000 transitions using 10 leapfrog steps per
 transition would take 0.35 seconds.
 Chain 1: Adjust your expectations accordingly!
 Chain 1:
 Chain 1:
 Chain 1: Iteration: 1 / 4000 [0%] (Warmup)
 Chain 1: Iteration: 400 / 4000 [10%] (Warmup)
 Chain 1: Iteration: 800 / 4000 [20%] (Warmup)
 Chain 1: Iteration: 1001 / 4000 [25%] (Sampling)
 Chain 1: Iteration: 1400 / 4000 [35%] (Sampling)
 Chain 1: Iteration: 1800 / 4000 [45%] (Sampling)
 Chain 1: Iteration: 2200 / 4000 [55%] (Sampling)
 Chain 1: Iteration: 2600 / 4000 [65%] (Sampling)
 Chain 1: Iteration: 3000 / 4000 [75%] (Sampling)
 Chain 1: Iteration: 3400 / 4000 [85%] (Sampling)
 Chain 1: Iteration: 3800 / 4000 [95%] (Sampling)
 Chain 1: Iteration: 4000 / 4000 [100%] (Sampling)
 Chain 1:
 Chain 1: Elapsed Time: 0.031 seconds (Warm-up)
 Chain 1: 0.089 seconds (Sampling)
 Chain 1: 0.12 seconds (Total)
 Chain 1:

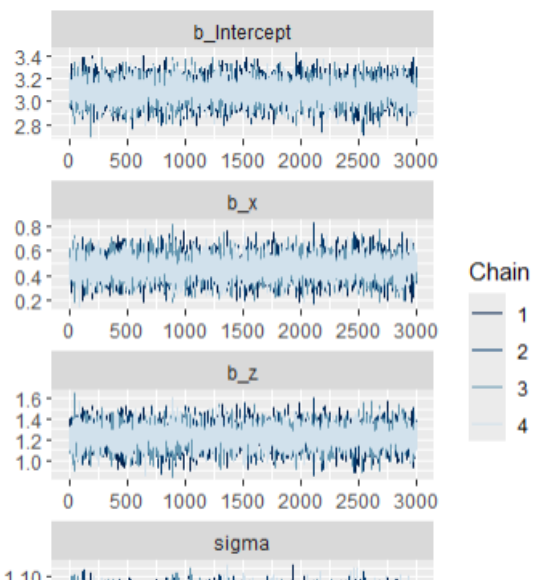
SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
 Chain 2:
 Chain 2: Gradient evaluation took 7.6e-05 seconds
 Chain 2: 1000 transitions using 10 leapfrog steps per
 transition would take 0.76 seconds.
 Chain 2: Adjust your expectations accordingly!
 Chain 2:
 Chain 2:
 Chain 2: Iteration: 1 / 4000 [0%] (Warmup)
 Chain 2: Iteration: 400 / 4000 [10%] (Warmup)
 Chain 2: Iteration: 800 / 4000 [20%] (Warmup)
 Chain 2: Iteration: 1001 / 4000 [25%] (Sampling)
 Chain 2: Iteration: 1400 / 4000 [35%] (Sampling)
 Chain 2: Iteration: 1800 / 4000 [45%] (Sampling)
 Chain 2: Iteration: 2200 / 4000 [55%] (Sampling)

Assessing the Training

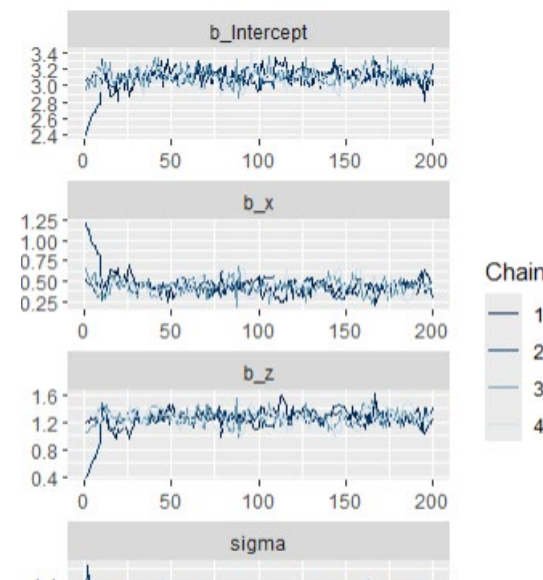
Bayesian data analysis excels at indicating when the training process has went wrong. To assess the training, there are several statistics we can consult, for example **visual assessment** (dubbed the “hairy caterpillar ocular inspection test” by Richard McElreath).



Positive visual assessment



Negative visual assesment



Assessing the Training

Another evaluation is the **Gelman-Rubin Convergence Statistic $\hat{R}$** calculated from the within- and between-chain variance. It quantifies how well *multiple chains have converged*.

```
R 4.5.0 using library(brms)
summary(model)
```

Regression Coefficients:

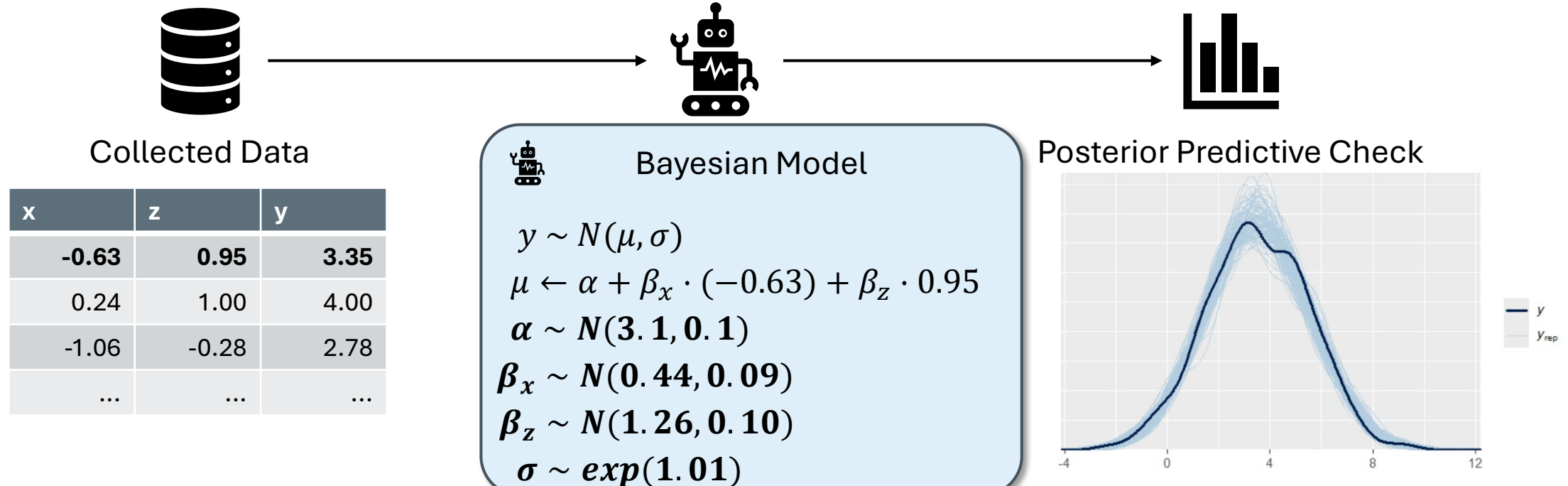
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	3.11	0.10	2.92	3.29	1.00	6767	7025
x	0.44	0.09	0.27	0.62	1.00	6066	6707
z	1.26	0.10	1.05	1.46	1.00	6011	6560

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	1.01	0.03	0.95	1.07	1.00	8808	6665

Assessing the Training

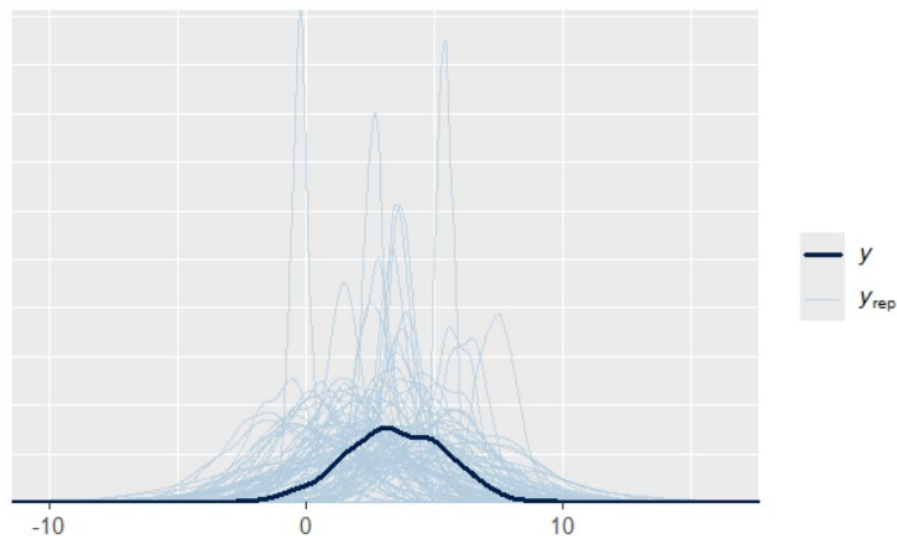
Similarly to the prior predictive check, we can run a **posterior predictive check** using the updated path coefficients.



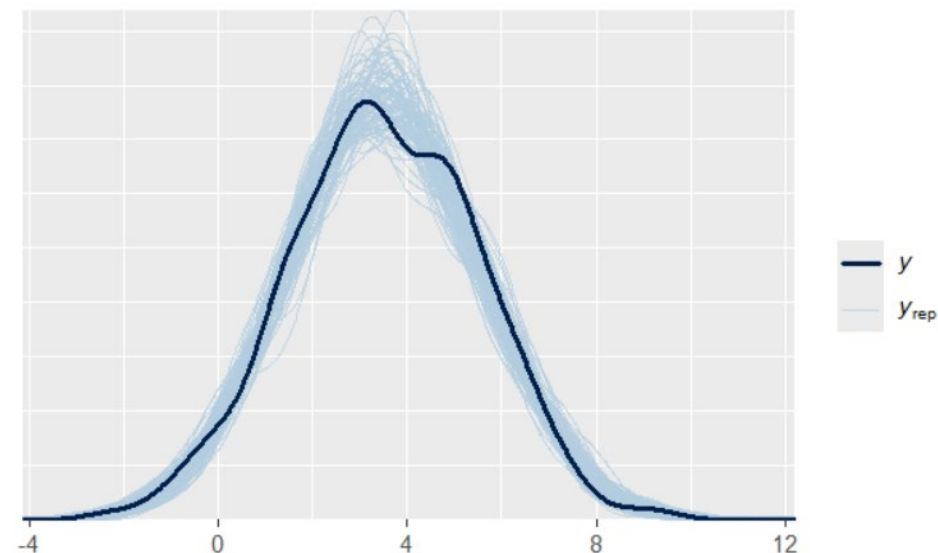
Assessing the Training

Comparing the prior with the posterior predictive check should show that the predictions of the model **grew closer** to the actually observed data.

Prior Predictive Check

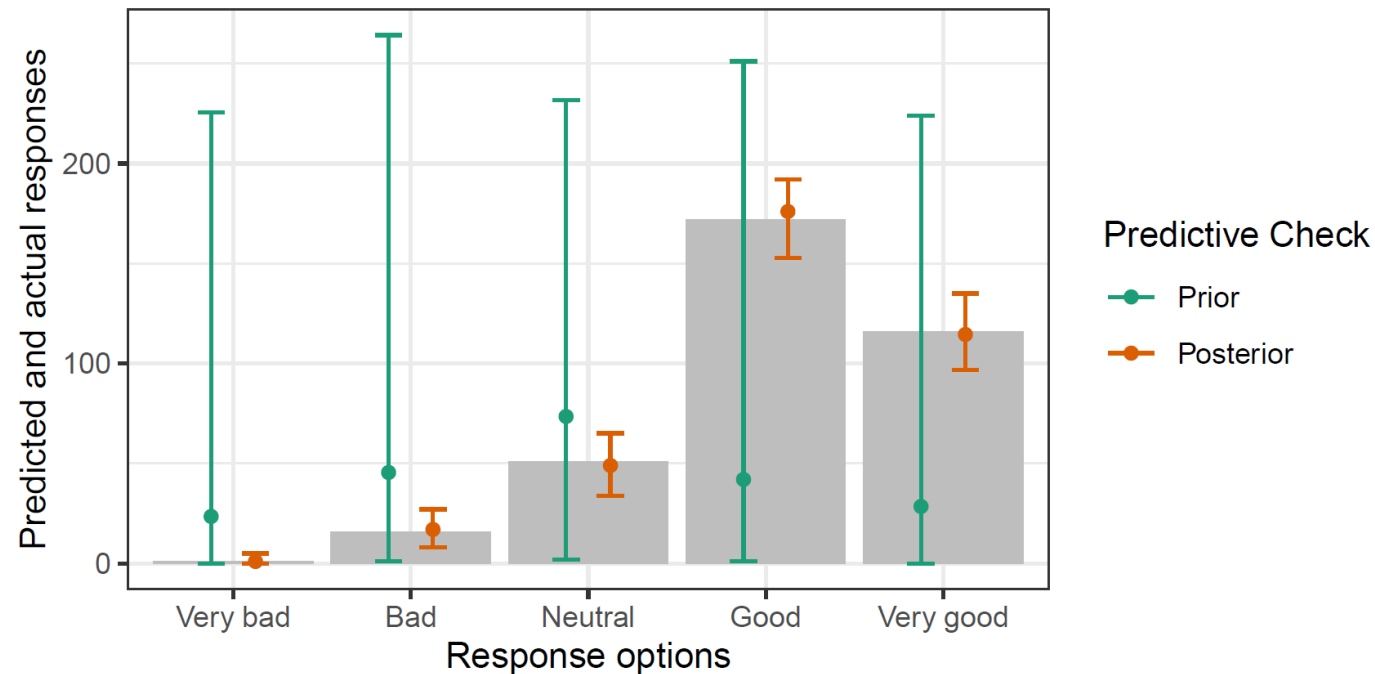


Posterior Predictive Check



Assessing the Training

Comparing the prior with the posterior predictive check should show that the predictions of the model **grew closer** to the actually observed data.



Interpreting the Results

After the Bayesian updating, the prior probability distributions have been updated to better reflect the observed data.

 Bayesian Model (prior)

$$y \sim N(\mu, \sigma)$$

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

$$\alpha \sim N(3, 1)$$

$$\beta_x \sim N(0, 0.5)$$

$$\beta_z \sim N(0, 0.5)$$

$$\sigma \sim \exp(1)$$

 Bayesian Model (posterior)

$$y \sim N(\mu, \sigma)$$

$$\mu \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

$$\alpha \sim N(\mathbf{3.1}, \mathbf{0.1})$$

$$\beta_x \sim N(\mathbf{0.44}, \mathbf{0.09})$$

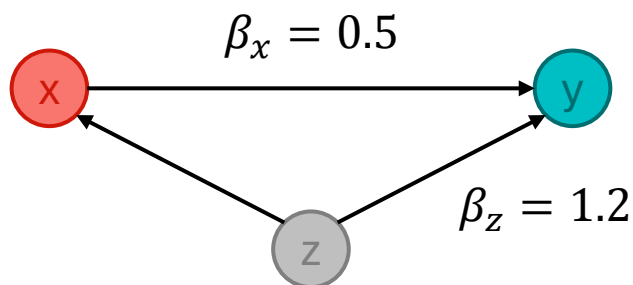
$$\beta_z \sim N(\mathbf{1.26}, \mathbf{0.10})$$

$$\sigma \sim \exp(\mathbf{1.01})$$

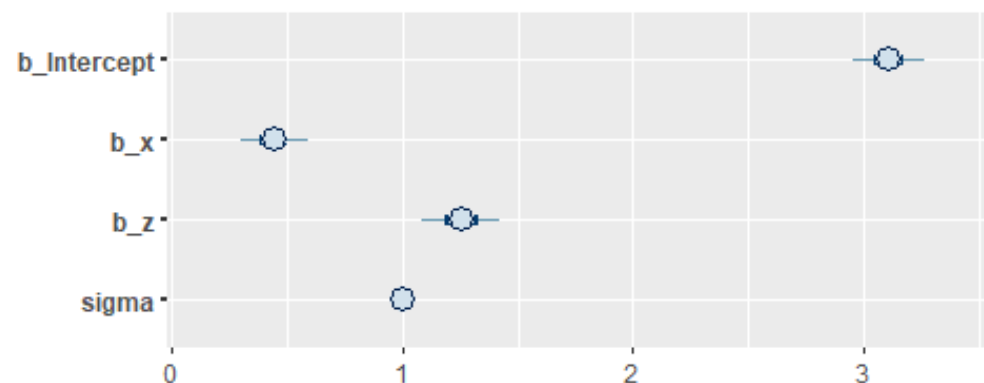
Interpreting the Results

After Bayesian updating, the posterior distributions represent the **effect that changing a variable** by one unit has on the (mean μ of the) outcome variable.

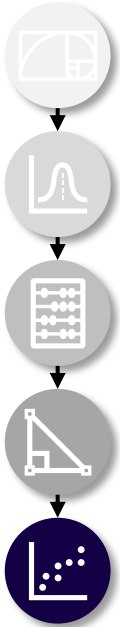
Simulated Effects



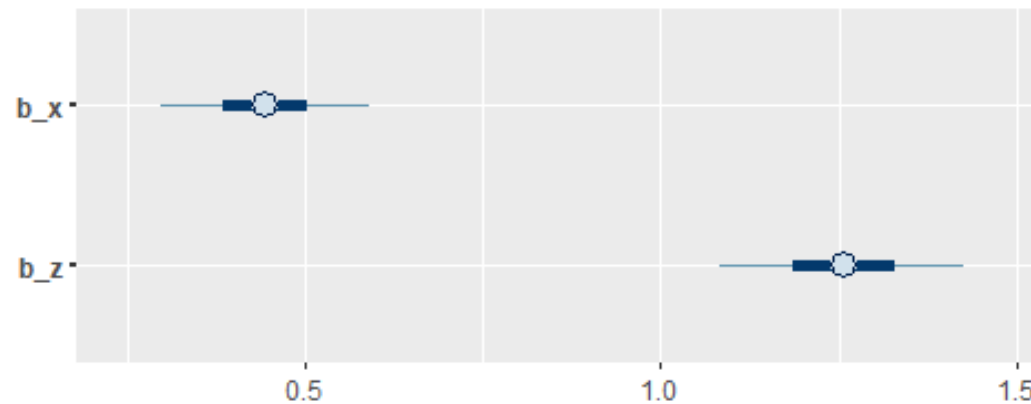
Estimated Effects



Credible Interval



The uncertainty around the estimation is a **credible interval**, not a *confidence interval*. It can be interpreted naturally: a 95% credible interval has a 95% chance of containing the true value of the parameter.



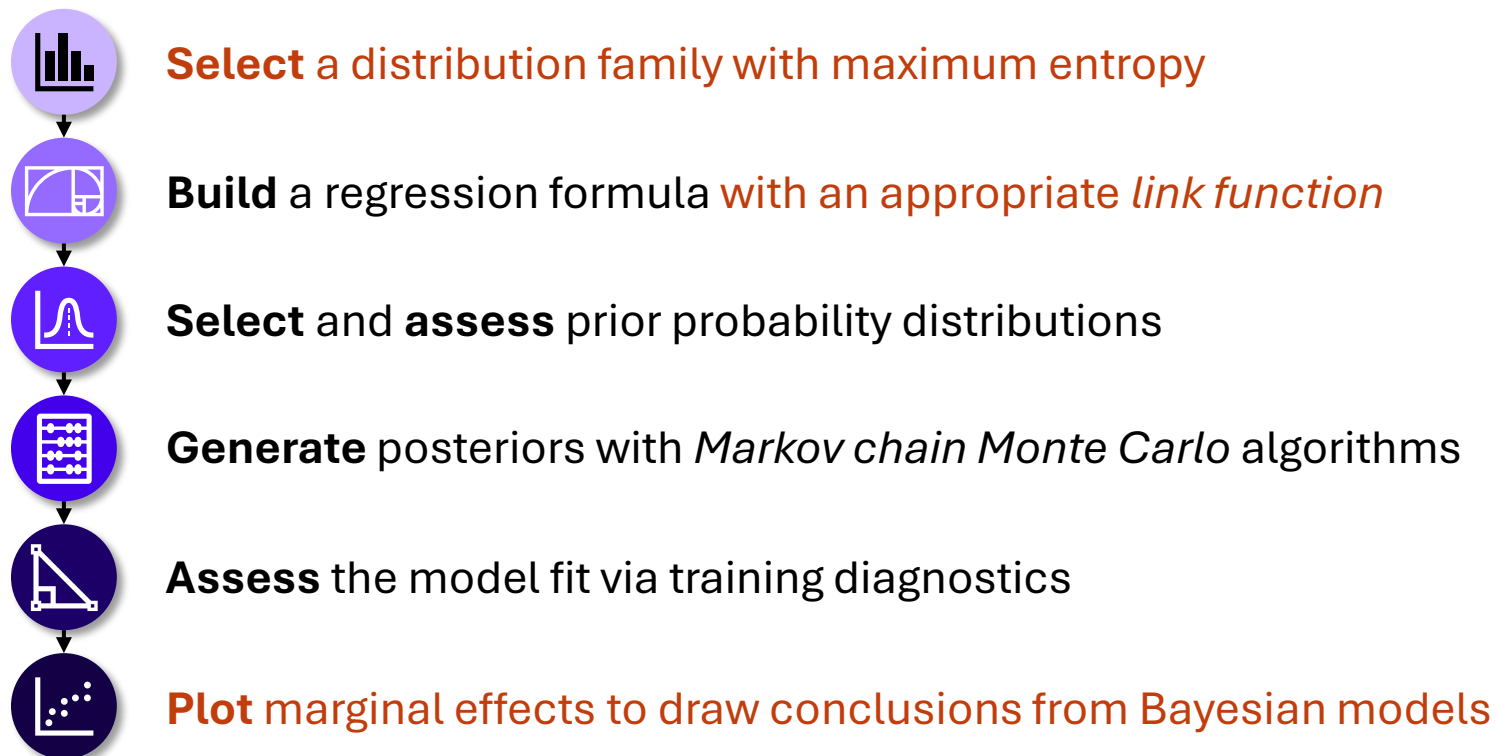
The further a CI is from 0.0, the *stronger* the effect. The narrower a CI, the *more certain* the effect.

Application II

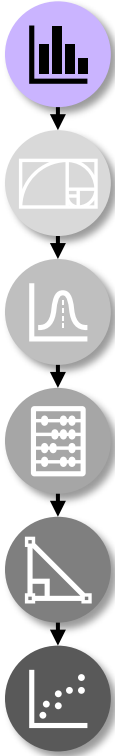
Live demonstration of a BDA workflow in practice

The Bayesian Workflow

The upcoming demonstration will teach you to *understand* or **do** the following **new** steps of a practical Bayesian workflow:



Normal Distribution



In Requirements and Software Engineering, **not many variables follow a normal** (i.e., Gaussian) **distribution**. Most frequentist data analyses only distinguishes between normal and non-normal distributions (via a simply Shapiro-Wilk test).



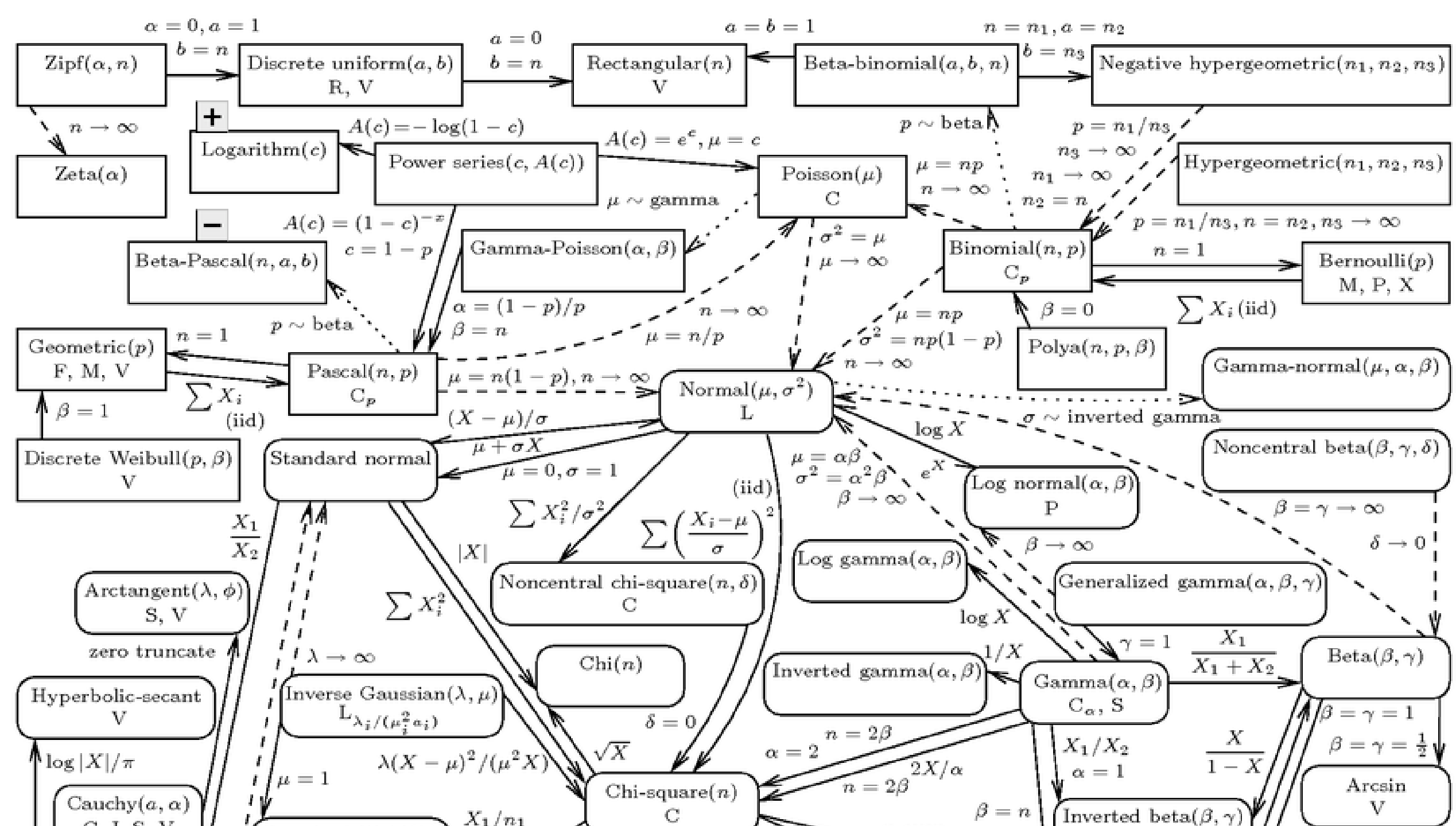
Gren, L., & Berntsson Svensson, R. (2021). Is it possible to disregard obsolete requirements? a family of experiments in software effort estimation. *Requirements Engineering*, 26(3), 459-480.

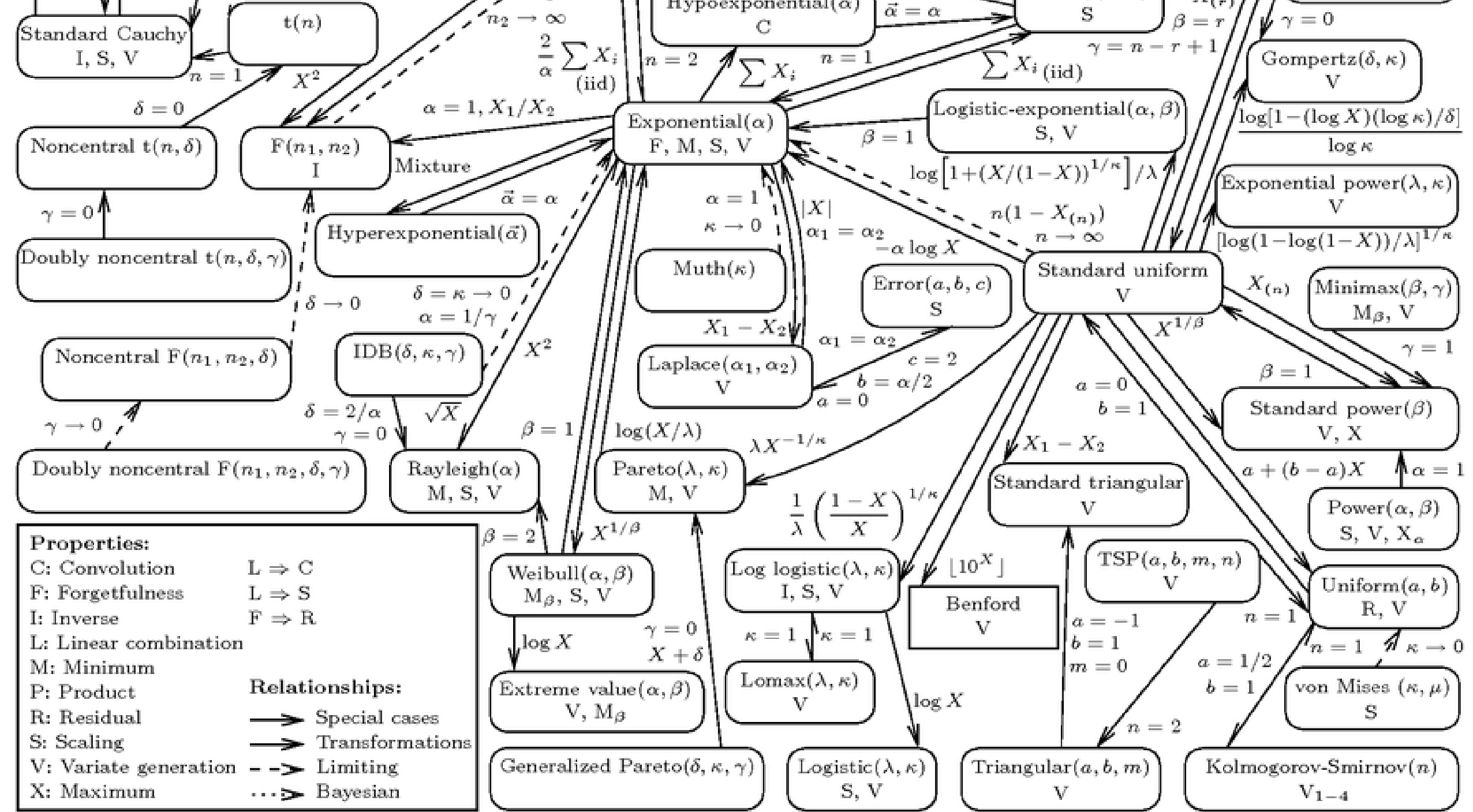
In most software engineering research, this distribution is often assumed to be normal (i.e. Gaussian), and if not, assumed to not exist and instead researchers use statistical tests based on ranks. However, **this is a pity since there are many probability distributions that could create much better models for the collected data.**

Besides the “normal” Gaussian, what other

Distributions

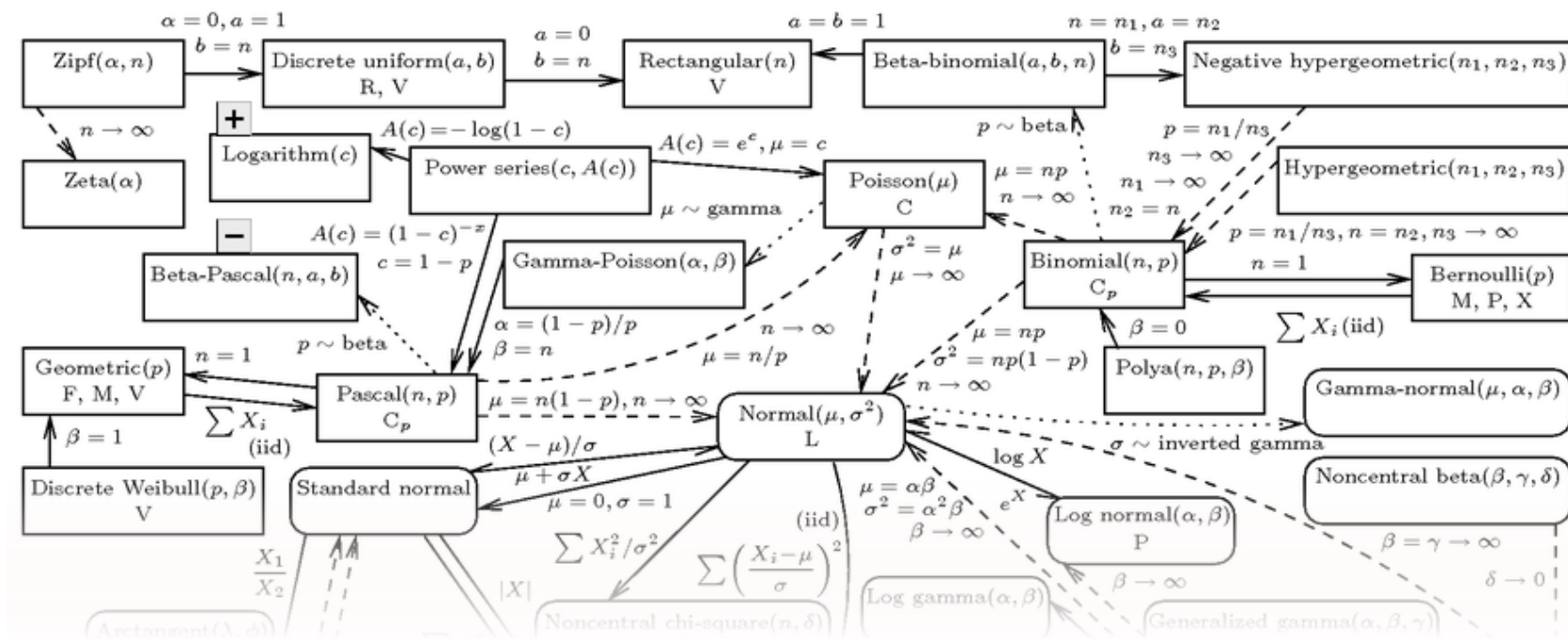
do you know?



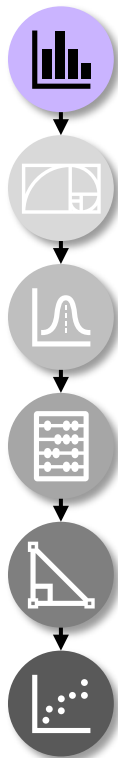


Distribution Families

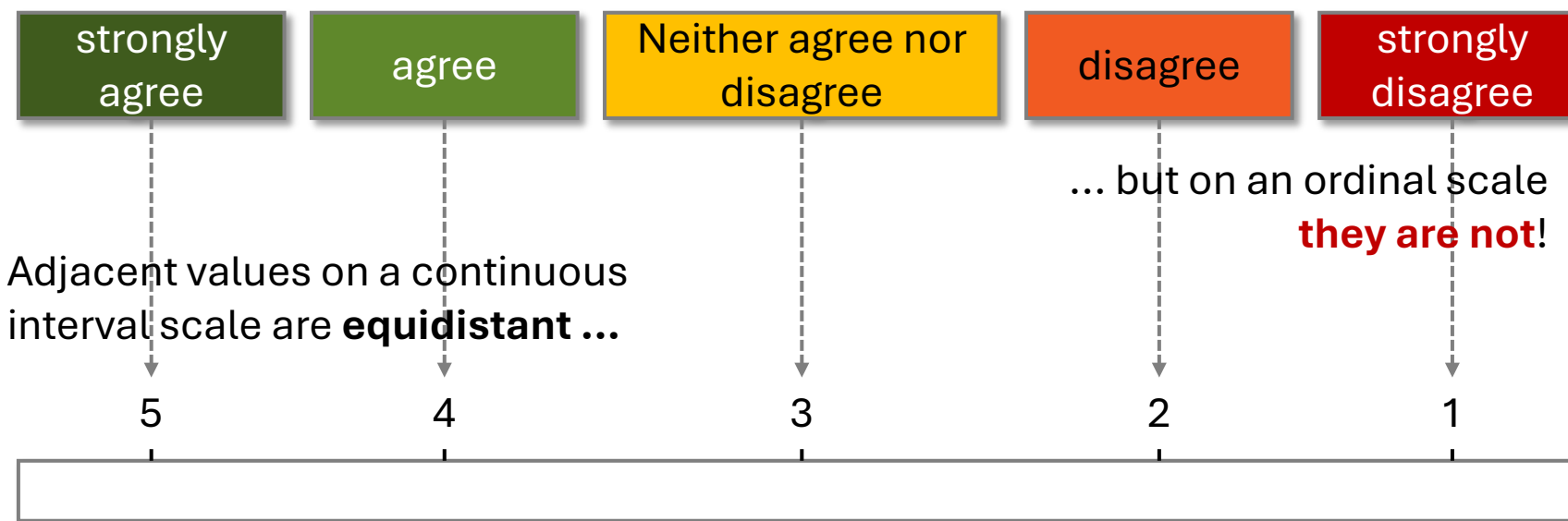
There are plenty of distribution families, each with its own assumption about how data is distributed



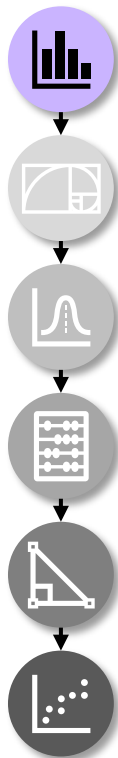
Classic Misconception



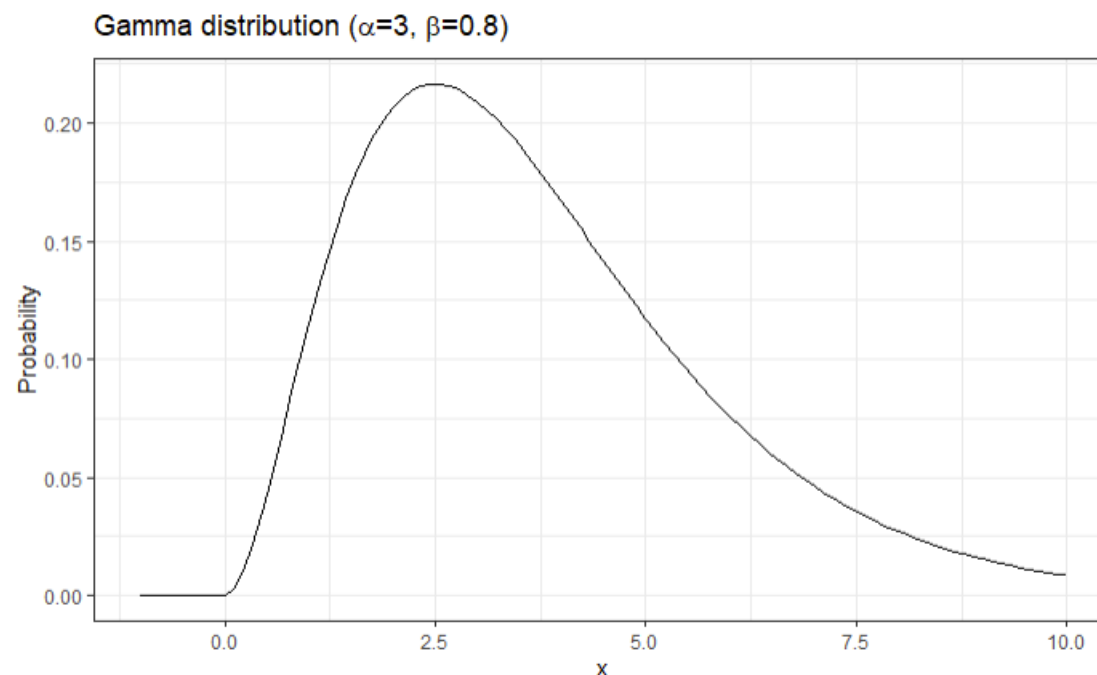
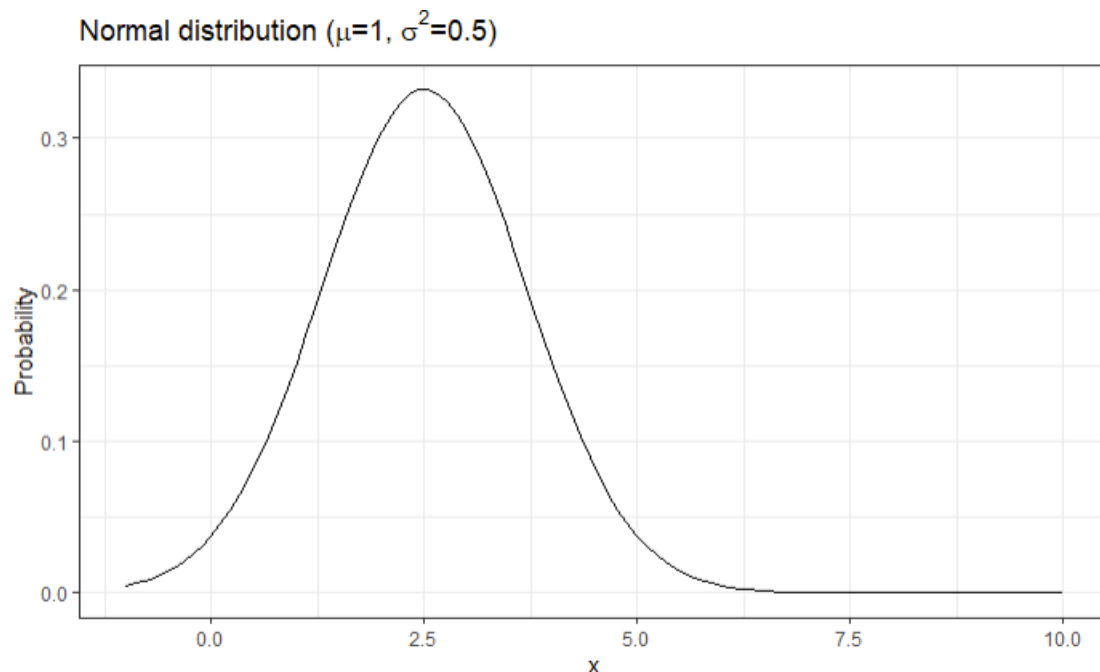
One of the most common forms of statistical malpractice is **treating ordinal data as continuous**.



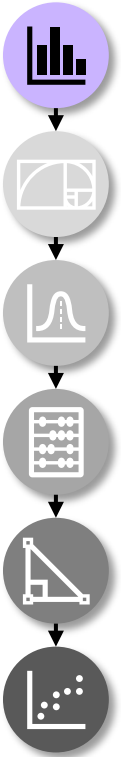
Picking the wrong Distribution



When we model our outcome variable with the wrong distribution family, we risk **omitting important properties** of the data.



Maximum Entropy



In order to decide, which distribution family to pick when modelling an outcome variable, we can use the **maximum entropy criterion**:

”

Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical review*, 106(4), 620.

[The maximum entropy distribution] is the least biased estimate possible on the given information; i.e., it is maximally noncommittal with regard to missing information.

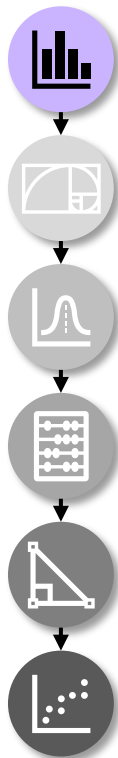
Among all probability distributions consistent with what you actually know about a variable, the "best" distribution to use is the one with **the largest entropy**, i.e., the one that is **maximally spread out** and **non-committal** beyond those constraints.

Maximum Entropy

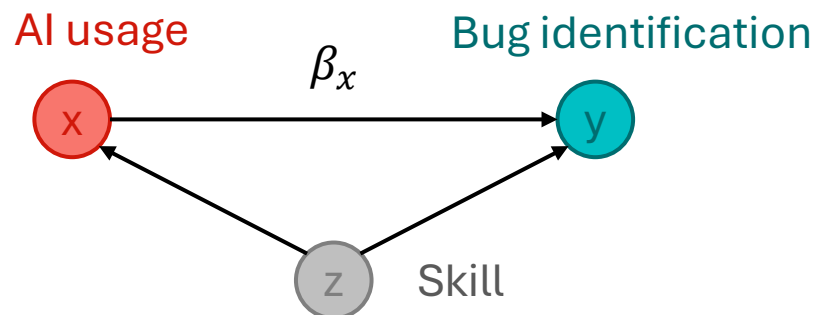
 Examples of the maximum entropy criterion:

- **Normal distribution:** continuous variable with fixed mean and variance
- **Binomial:** fixed number of trials and probability of success
- **Beta-binomial:** fixed number of trials but *varying probability of success*
- **Poisson:** unbounded count with fixed mean at a constant rate
- **Pascal:** unbounded count with fixed mean but *at a non-constant rate* (i.e., with a larger variance)

Updated Context



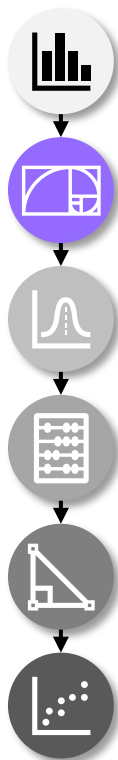
Considering the same research question, but now measuring the **outcome variable y** as the *number of times a code reviewer found the one bug in a code snippet*.



Var	Name	Type
x	Measure of AI usage	Continuous, Gaussian-distributed
y	Number of code snippet where a bug was identified	Positive count of successful trials
z	Skill of a code reviewer	Continuous, Gaussian-distributed

The **binomial distribution** has maximum entropy under these constraints.

Building a Regression Formula



Modelling the outcome variable y with the Binomial distribution requires updating our Bayesian model:



Bayesian Model

$$\begin{aligned}
 y &\sim \text{Binomial}(n, p) \\
 p &\leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z \\
 \alpha &\sim N(0, 1) \\
 \beta_x &\sim N(0, 0.5) \\
 \beta_z &\sim N(0, 0.5)
 \end{aligned}$$

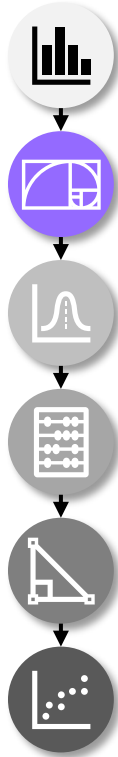
R 4.5.0 using library(brms)

```

model <- brm(
  y | trials(n) ~ 1 + x + z,
  family = binomial,
  priors = c(
    prior(normal(0, 1), class = Intercept),
    prior(normal(0, 0.5), class = b)),
  iter = 4000, warmup = 1000, chains = 4,
  cores = 4, seed = 4
)

```

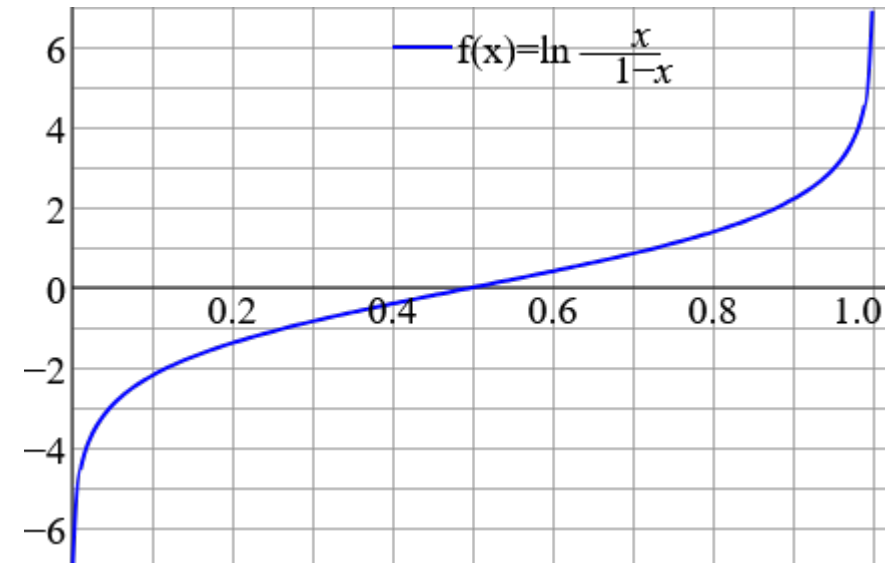
Link Functions



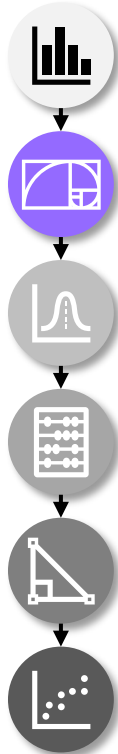
Some probability distribution families have specific requirements for their parameters (e.g., $p \in [0, 1]$). Still, we would like to use “natural” linear models in our models. As a compromise, we can use **link functions**.

$$y \sim \text{Binomial}(n, p)$$

$$\text{logit}(p) \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$



Bayesian Model



The link function must be specified in a mathematical model, but is included in `brms` by default.



Bayesian Model

$$\begin{aligned}
 y &\sim \text{Binomial}(n, p) \\
 \text{logit}(p) &\leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z \\
 \alpha &\sim N(0, 1) \\
 \beta_x &\sim N(0, 0.5) \\
 \beta_z &\sim N(0, 0.5)
 \end{aligned}$$

R 4.5.0 using `library(brms)`

```

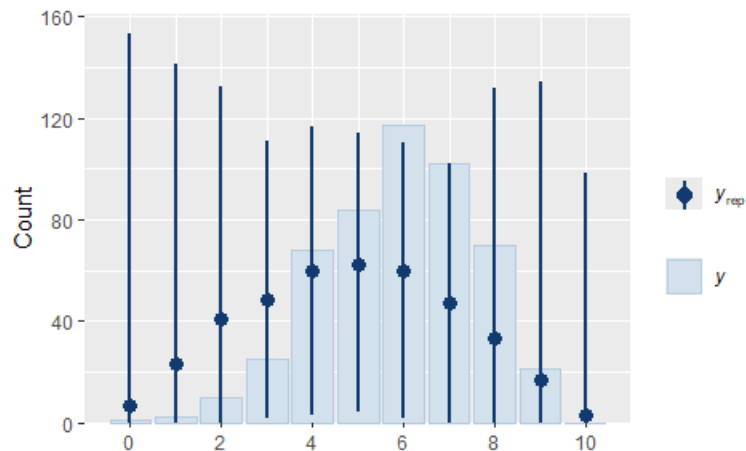
model <- brm(
  y | trials(n) ~ 1 + x + z,
  family = binomial(link="logit"),
  priors = c(
    prior(normal(0, 1), class = Intercept),
    prior(normal(0, 0.5), class = b)),
  iter = 4000, warmup = 1000, chains = 4,
  cores = 4, seed = 4
)

```

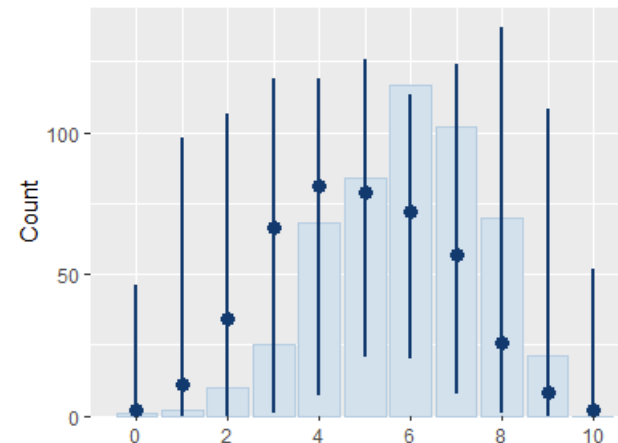
Prior Predictive Checks, Training, and Posterior Predictive Checks

The link function complicates the selection of appropriate priors and makes prior predictive checks all-the-more-important.

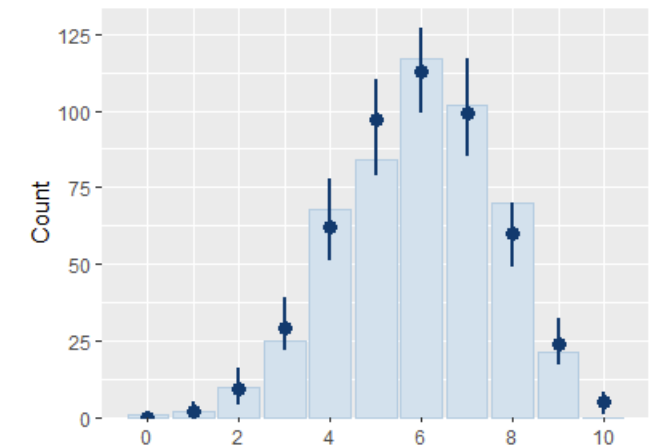
 **Suboptimal** prior predictive check with
`prior(normal(0, 0.5), class = b)`



 **Better** prior predictive check with
`prior(normal(0, 0.3), class = b)`



Posterior Predictive Check



Model Interpretation

Due to the link function, it is no longer trivial to interpret the posterior distributions.

```
R 4.5.0 using library(brms)
summary(model)
```

Regression Coefficients:							
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.27	0.07	0.13	0.40	1.00	4468	5088
x	0.11	0.06	-0.01	0.23	1.00	4435	4828
z	0.20	0.07	0.06	0.33	1.00	4576	5358



Bayesian Model

$$y \sim \text{Binomial}(n, p)$$

$$\text{logit}(p) \leftarrow \alpha + \beta_x \cdot x + \beta_z \cdot z$$

$$\alpha \sim N(0, 1)$$

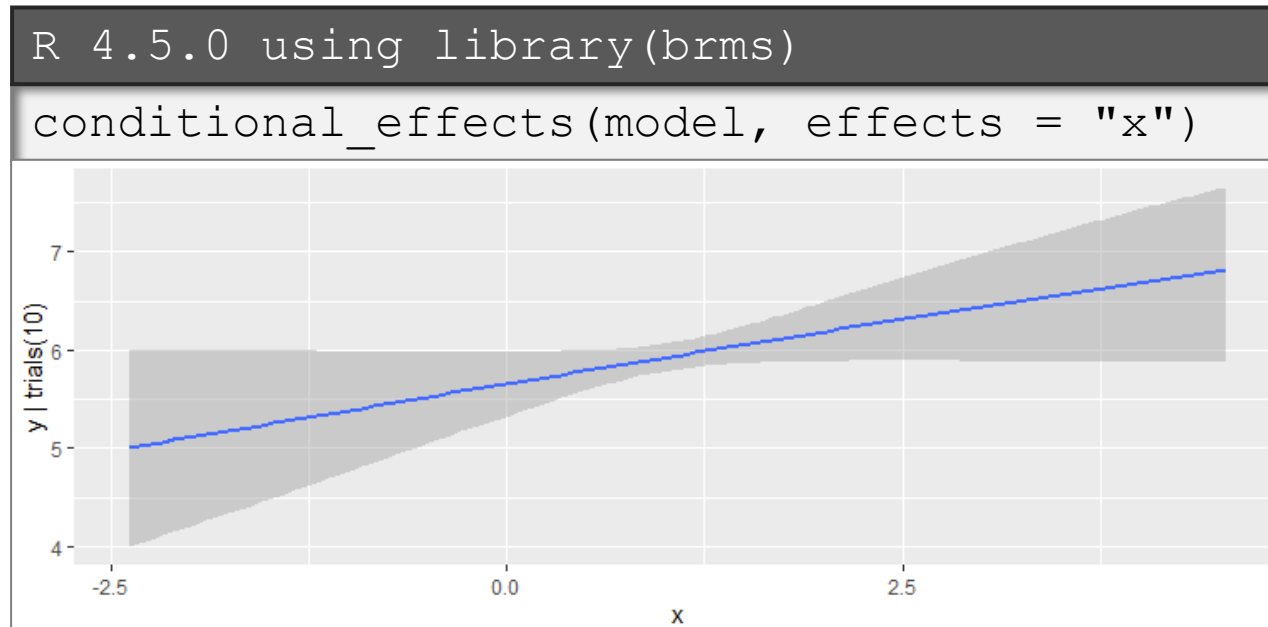
$$\beta_x \sim N(0, 0.3)$$

$$\beta_z \sim N(0, 0.3)$$

Instead, we can plot **marginal effects**.

Marginal Effects

A marginal effect is the effect of one variable on the outcome while holding **all other variables constant** at representative levels.



Marginal Effects

A marginal effect is the effect of one variable on the outcome while holding **all other variables constant** at representative levels.

What is your way-of-working in your project?

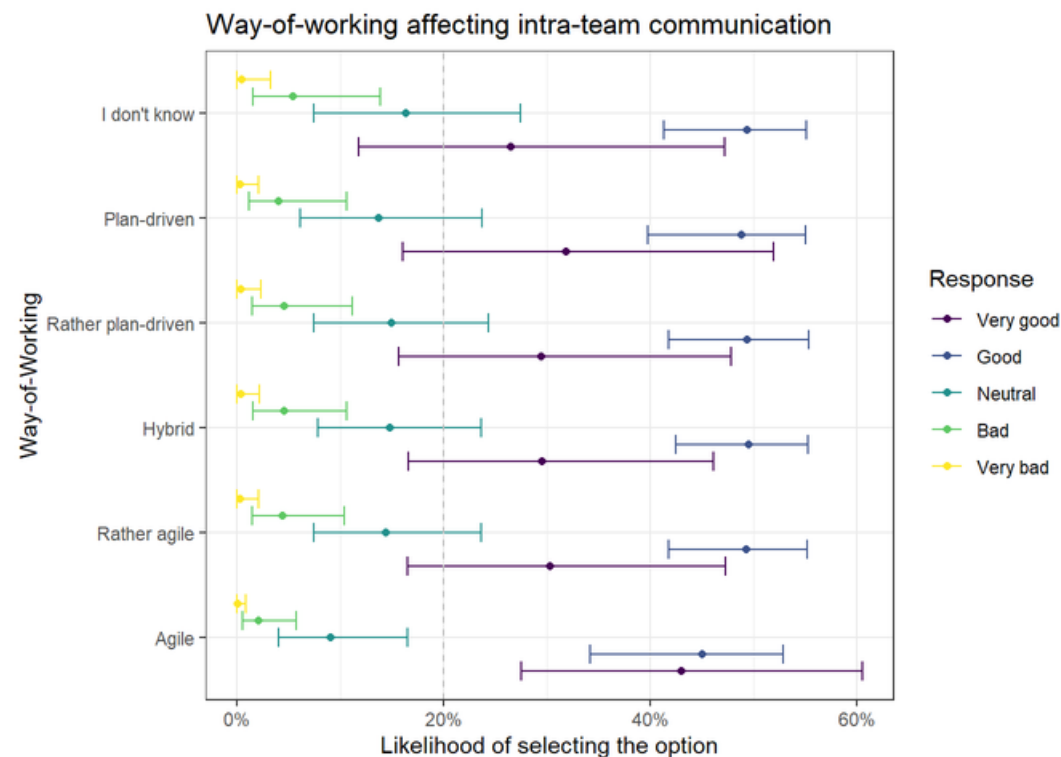
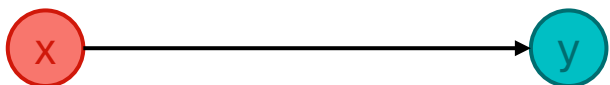
Plan-driven	Rather plan-driven	Hybrid	Rather agile	Agile
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How do you perceive the communication inside your team?

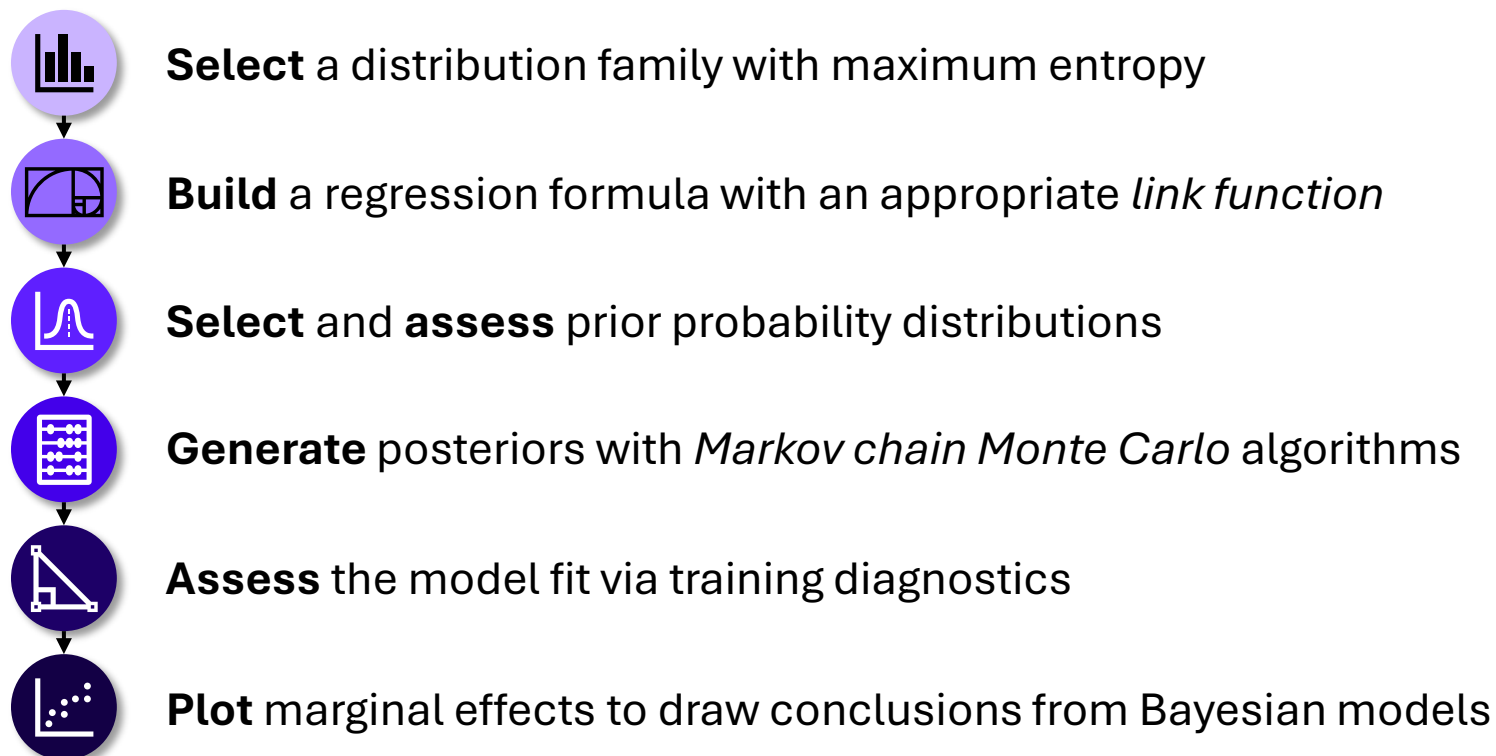
Very bad	Bad	Neutral	Good	Very Good
----------	-----	---------	------	-----------

way-of-working

Intra-team communication



The Bayesian Workflow



Path Ahead

BDA in Action

We have successfully published studies using BDA in RE/SE venues.

the Bayesian data analysis methods, we first selected an appropriate distribution type for each response variable [58]. We selected the distribution types based on the maximum entropy criterion [59] and ontological assumptions. As the values of all metrics are bounded within $[0, 1]$, we chose a distribution from the Beta family [60]. The Beta distribution family deals with the heteroskedasticity of the data at the edges of the $[0, 1]$ range. To account for occurrences of the values 0 and 1 in each metric, we selected the *zero-one-inflated Beta distribution* [61].

Next, we defined regression formulae for each response variable. These specify each response variable in relationship to all available independent variables. The strength and direction of the effect of each variable—the quantity of interest to answer our research question—is represented by a coefficient in the form of a random variable.

We selected uninformative prior distributions for the coefficients of each included predictor and confirmed their eligibility via prior predictive checks [62]. After confirming their eligibility, we trained each model with the collected data. Hamiltonian Monte Carlo Markov Chains (MCMC) [63] update the coefficient distributions based on the empirical data. During this process, the parameters of the coefficient distributions are adjusted to better predict the response variable based on the independent variables [51]. After the training process, we perform posterior predictive checks, which work similarly to the prior predictive check but use the updated posterior coefficient distributions instead of the prior distributions.

To evaluate each model, we inspected the distributions of the updated path coefficients. Traditionally, coefficients where the 95% credibility interval (CI) is consistent with 0, i.e., where the CI overlaps with 0, are not considered statistically significant [55]. To avoid the fallacy of reducing complex data down to the binary property of statistical significance, we still investigate all factors whose coefficient shows a notable effect. Therefore, in addition to the statistically significant factors we consider coefficients where the 50% credibility interval is not consistent with 0 to show a weak impact.

While the posterior path coefficients answer RQ1, their interpretation is challenging since the regression formulae are wrapped by a *logit-link* function [51]. This link function scales the summed value of all regressors to the range $(0, 1)$, which is the appropriate range for the estimated main param-

interaction effect between this factor and the TLR approach. Such an interaction effect discerns whether a quality factor affects the response variable differently for different TLR approaches [65]. We only used the F₂-score as a response instead of all four performance metrics to limit the space of potential models. We plotted the conditional effects to visualize similar and different effects of quality factors across TLR approaches, which provides insight into whether some approaches deal with certain significant factors better or worse than others.

IV. RESULTS

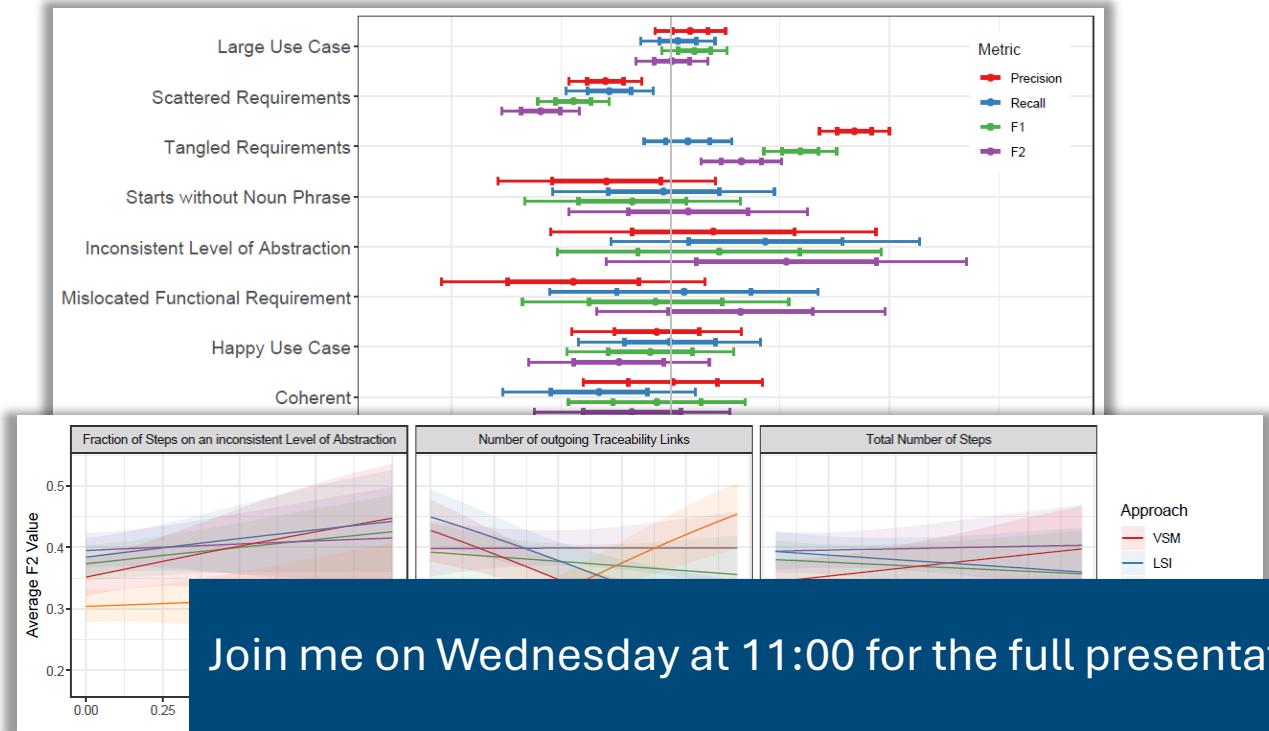
In this section, we show the results from the data analysis and set them into context of existing requirements quality work. For conciseness, we report only relevant and meaningful results. Our replication package contains the data, material, and figures of all omitted results [45].

A. Posterior Coefficient Distributions

Figure 3 visualizes the posterior distributions of variable coefficients from the data analysis that were at least weakly significant. The dots represent the average effect of the factor on the respective response variable (coded in colors), while the two whiskers represent the single and double standard deviation distance from the average. The further the dot is away from 0, the stronger the effect. The more narrow the whiskers, the more certain the effect.

Of the 16 investigated quality factors, 13 show at least a weakly significant effect on at least one of the four outcome metrics. Some factors exhibit narrow CIs, suggesting a very certain estimated effect. *Large Use Cases* affect only precision and the F₁-score slightly, where *Scattered Requirements* impede all metrics significantly. *Tangled Requirements*, on the other hand, have a positive effect on all metrics but recall.

The other factors have less certain effects on the response variables. Notable are the negative effects of use cases with a high *Requirements Length*, and the ratio of sentences that contain *Anaphora* and *Negatives* on the performance of TLR approaches. On the other hand, the degree to which a requirement is written on an *Inconsistent Level of Abstraction* and *Contains Alternatives* will benefit particularly recall and F₂-score. Sentence-level complexity significantly benefits precision at a minor cost to recall, improving F₁- and F₂-scores



Hey, T. & Frattini, J. (2026). How Requirements Quality Makes (or Breaks) Traceability Link Recovery. In 2026 IEEE 34th International Requirements Engineering Conference (RE). IEEE. <https://doi.org/10.48550/arXiv.2606.11834>

Learning Bayesian Data Analysis

If you have (1) a causal hypothesis and (2) quantitative data about its variables, you can start on your own BDA path by:

1. **Setting up Stan:** The computational engine behind contemporary BDA approaches is Stan (<https://mc-stan.org/>). Most programming languages have an interface for it (e.g., `brms` in R)
2. **Testing the Bayesian workflow:** Follow the basic BDA workflow presented in this tutorial to derive inferential conclusions from raw data.
3. **Learn distributions:** The full potential of BDA is unlocked when mastering different distributions to represent response variables.

Distributions

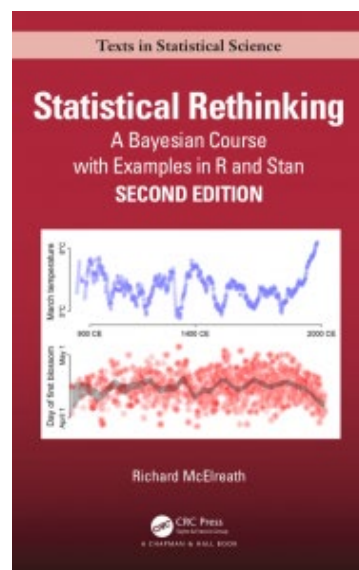
When learning distributions in addition to the normal distribution, consider the following:

Name	Description	Examples
Binomial	n independent trials with a success rate of p for each trial	Identifying each bug in n codes snippets
Poisson/Pascal	Discrete count of events	Count of identified bugs
Categorical	Extension of the binary response with k independent categories	Single-choice response
Ordinal	Categorical data with a linear order between levels	Likert-scale response
Beta	Continuous data bounded between $[0; 1]$	F_β -score metrics

Resources

The following resources are the state-of-the-art for learning and applying BDA in practice:

The Book



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CRAN

R Project

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The R Project for Statistical Computing

Getting Started

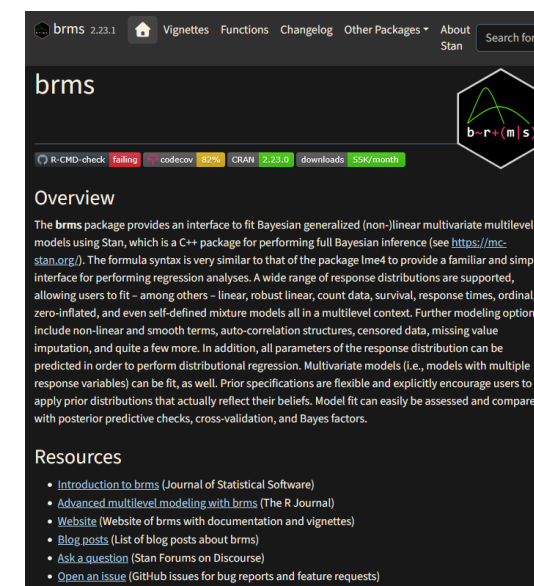
R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred [CRAN mirror](#).

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- **R version 4.6.1 (Happy Hop)** has been released on 2026-06-24.
- **R version 4.5.3 (Reassured Reassurer)** (wrap-up release) was released on 2026-03-11.
- The **user! 2026** conference took place in Warsaw, Poland, July 7-9.
- You can support the R Foundation with a renewable subscription as a [supporting member](#).

The Library



Resources

Other useful resources:

- Blog by Andrew Heiss: <https://www.andrewheiss.com/blog/>
- Blog by Solomon Kurz: <https://solomonkurz.netlify.app/blog/>
- Follow Richard Torkar and Robert Feldt

...

Opportunities

Beyond the content of this tutorial, BDA has many more advantages that all naturally embed into the presented workflow, like modelling

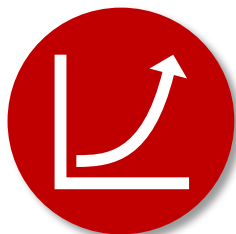
- **missing data**, e.g., when the data could not be collected or you have partial responses to a survey form,
- **mixed distributions**, e.g., hurdle models describing a data-generation process stemming from two separate processes,
- **censored or truncated data**, e.g., when you have to cut off participants of an experiments as they run out of time,
- **hierarchical effects**, e.g., when you have repeated measures per subject of your study, or
- **interaction effects**, e.g., when an external variable moderates the effect of the main cause on the main effect.

Conclusion

Reflection, discussion, and closing

Bayesian Data Analysis in a Nutshell

The Bayesian paradigm provides an alternative to the common frequentist paradigm for data analysis.



Learning Curve

The learning curve is steep and you need to pick up several new skills



Process

Thankfully, guidance and tooling for BDA got better



Transparent & Sophisticated

BDA rewards you with explicit, traceable, and interpretable results

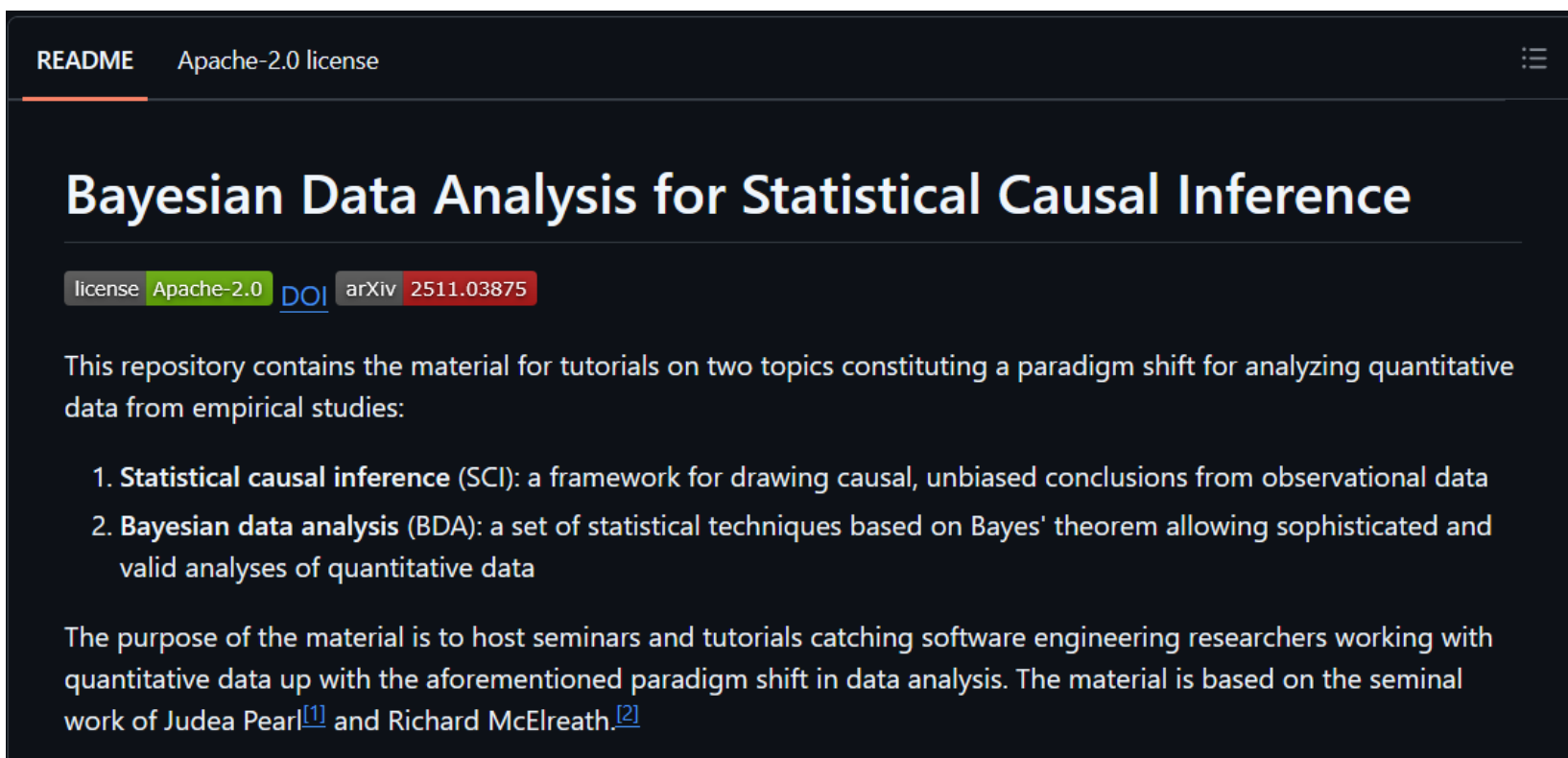


Perfect Timing

BDA is up-and-coming in SE/RE research, so you are just in time to join and learn

Material

You find all material online in our public repository at
<https://github.com/JulianFrattini/bda4sci>



The screenshot shows the GitHub repository page for 'Bayesian Data Analysis for Statistical Causal Inference'. The page has a dark theme. At the top, there is a navigation bar with 'README' and 'Apache-2.0 license'. Below this, the repository title is displayed in large white text. Under the title, there are badges for 'license Apache-2.0', 'DOI', and 'arXiv 2511.03875'. The main content area contains a paragraph describing the repository's purpose and a numbered list of two topics: 'Statistical causal inference (SCI)' and 'Bayesian data analysis (BDA)'. At the bottom, there is a paragraph explaining the purpose of the material and mentioning Judea Pearl and Richard McElreath with links to their work.

README Apache-2.0 license

Bayesian Data Analysis for Statistical Causal Inference

license Apache-2.0 DOI arXiv 2511.03875

This repository contains the material for tutorials on two topics constituting a paradigm shift for analyzing quantitative data from empirical studies:

1. **Statistical causal inference (SCI)**: a framework for drawing causal, unbiased conclusions from observational data
2. **Bayesian data analysis (BDA)**: a set of statistical techniques based on Bayes' theorem allowing sophisticated and valid analyses of quantitative data

The purpose of the material is to host seminars and tutorials catching software engineering researchers working with quantitative data up with the aforementioned paradigm shift in data analysis. The material is based on the seminal work of Judea Pearl^[1] and Richard McElreath.^[2]

Let's go improve

Conclusion Validity

with the help of

Bayesian Data Analysis